PART II

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13 Structure and Functions of the Dentin-Pulp Complex

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CHAPTER OUTLINE

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Morphologic Zones of the Pulp

THE PULP-DENTIN COMPLEX

The dental pulp and dentin function as a unit and the odontoblasts represent a crucial element in this system. The odontoblasts are located in the periphery of the pulp tissue, with extensions into the inner part of dentin. Dentin would not exist unless produced by odontoblasts, and the dental pulp is dependent upon the protection provided by the dentin and enamel. Likewise, integrated dynamics of the pulp-dentin complex imply that impacts on dentin may affect the pulpal components and that disturbances in the dental pulp will in turn affect the quantity and quality of the dentin produced.

ODONTOBLAST LAYER

The outermost stratum of cells of the healthy pulp is the odontoblast layer (Figs. 13.1 and 13.2). This layer is located immediately subjacent to the predentin. The odontoblast processes, however, pass on through the predentin into the inner part of dentin. Consequently, the odontoblast layer is actually composed of the cell bodies of odontoblasts. In addition, capillaries, nerve fibers, and dendritic cells may be found among the odontoblasts.

In the coronal portion of a young pulp that is actively secreting collagen, the odontoblasts assume a tall columnar form.[63] The odontoblasts vary in height; consequently, their nuclei are not all at the same level and are aligned in a staggered array, often described as a palisade appearance. This organization makes the layers appear to be three to five cells in thickness even though there is only one actual layer of odontoblasts. Between adjacent odontoblasts there are small intercellular spaces approximately 30 to 40 nm in

*The authors acknowledge the outstanding work of Drs. Henry Trowbridge, Syngcuk Kim, Hideaki Suda, David H. Pashley, and Fredrick R. Liewehr in previous editions of this text. The present chapter is built on their foundational work.

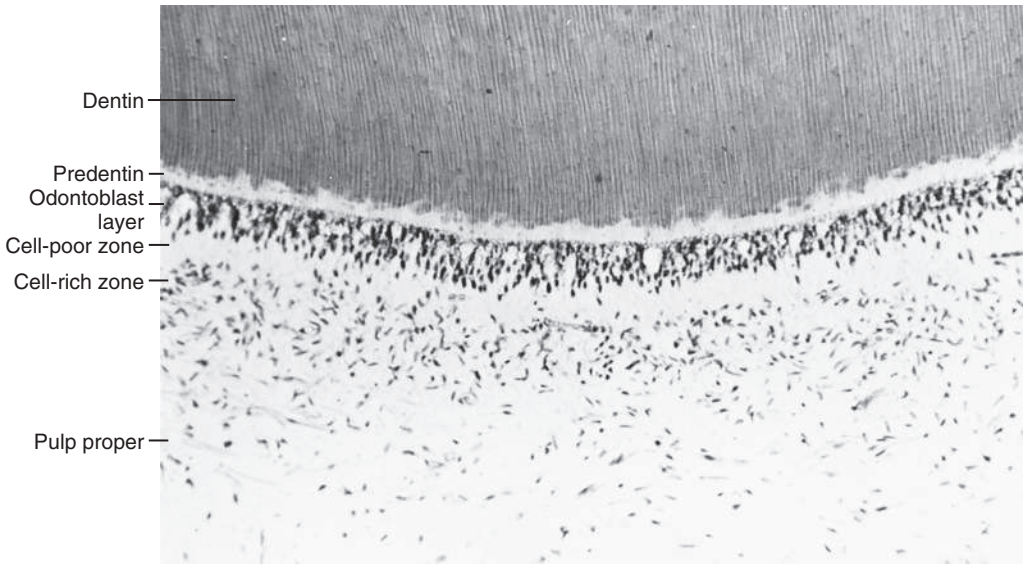


Fig. 13.1 Morphologic zones of the mature pulp.

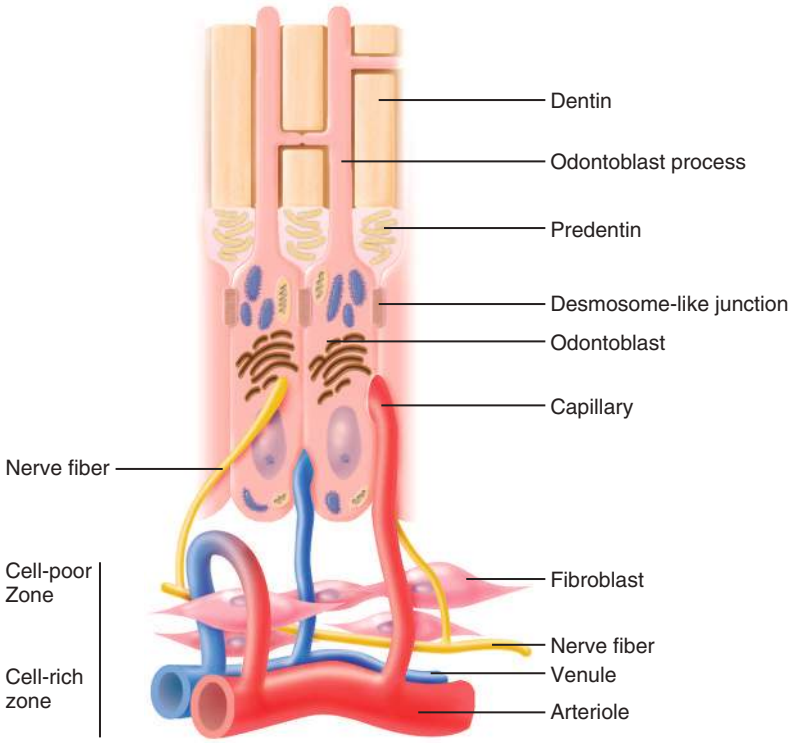


Fig. 13.2 Diagrammatic representation of the odontoblast layer and subodontoblastic region of the pulp.

width. Odontoblast cell bodies are connected by tight and gap junctional complexes.[29,63,162] Gap junctions are formed by connexin proteins[115] that permit cell-to-cell passage of signal molecules.

The odontoblast layer in the coronal pulp contains more cells per unit area than in the radicular pulp.[245] Whereas the odontoblasts of the mature coronal pulp are usually columnar, those in the midportion of the radicular pulp are more cuboidal.[63] Near the apical foramen, the odontoblasts appear as a squamous layer of flattened cells. Because fewer dentinal tubules per unit area are present in the root than in the crown of the tooth, the odontoblast cell bodies are less crowded and are able to spread out laterally.[245] During maturation and aging, there is a continued ongoing crowding in the odontoblast layer, particularly in the coronal pulp, due to narrowing of the pulp space. Apoptosis of odontoblasts seems to adjust for this limited space during development.[257]

There is a series of specialized cell-to-cell junctions (i.e., junctional complexes), including desmosomes (i.e., zonula adherens), gap junctions (i.e., nexuses), and tight junctions (i.e., zonula occludens) that connect adjacent odontoblasts.

Spot desmosomes located in the apical part of odontoblast cell bodies mechanically join odontoblasts together. The numerous gap junctions provide permeable pathways through which signal molecules can pass between cells (Fig. 13.3) to synchronize secretory activity that produces relatively uniform predentin layers (see Fig. 13.2). These junctions are most numerous during the formation of primary dentin. Gap junctions and desmosomes have also been observed joining odontoblasts to the processes of fibroblasts in the subodontoblastic area. Tight junctions are found mainly in the apical part of odontoblasts in young teeth. These structures consist of linear ridges and grooves that close off the intercellular space. However, tracer studies suggest direct passage of small elements from subodontoblastic capillaries to predentin and dentin between the odontoblasts.[359] It appears that tight junctions determine the permeability of the odontoblast layer when dentin is covered by enamel or cementum by restricting the passage of molecules, ions, and fluid between the extracellular compartments of the pulp and predentin.[29] During cavity preparation, these junctions are disrupted, thereby increasing dentin permeability.[380,381]

CELL-POOR ZONE

Immediately subjacent to the odontoblast layer in the coronal pulp, there is often a narrow zone approximately 40 μm in width that is relatively free of cells (see Fig. 13.1) and hence is called the *cell-free layer of Weil*. It is traversed by blood capillaries, unmyelinated nerve fibers, and the slender cytoplasmic processes of fibroblasts (see Fig. 13.2). The presence or absence of the cell-poor zone depends on the functional status of the pulp.[63] It may not be apparent in young pulps, where dentin forms rapidly, or in older pulps, where reparative dentin is being produced.

CELL-RICH ZONE

In the subodontoblastic area, there is a stratum containing a relatively high proportion of fibroblasts compared with the more central region of the pulp (see Fig. 13.1). It is much more prominent in the coronal pulp than in the radicular pulp. Besides fibroblasts, the cell-rich zone may include a variable number of immune cells like macrophages and dendritic cells but also undifferentiated mesenchymal stem cells (MSCs).

On the basis of evidence obtained in rat molar teeth, it has been suggested[128] that the cell-rich zone forms as a result of peripheral migration of cells populating the central regions of the pulp, commencing at about the time of tooth eruption. Migration of immunocompetent cells out of and into the cell-rich zone has been demonstrated as a result of antigenic challenge.[428] Although cell division within the cell-rich zone is rare in normal pulps, the death of the odontoblasts triggers a great increase in the rate of mitosis. Because odontoblasts are postmitotic cells, irreversibly injured odontoblasts are replaced by cells that migrate from the cell-rich zone onto the inner surface of the dentin.[100] This mitotic activity is probably the first step in the formation of a new odontoblast layer.[74,261,263,345] Studies implicate stem cells as a source for these replacement odontoblasts.[353]

PULP PROPER

The pulp proper is the central mass of the pulp (see Fig. 13.1). It consists of loose connective tissue and contains the larger blood vessels and nerves. The most prominent cell in this zone is the fibroblast.

Cells of the Pulp

ODONTOBLAST

Because odontoblasts are responsible for dentinogenesis, both during tooth development and aging, the odontoblast is the most characteristic and specialized cell of the dentin-pulp complex. During dentinogenesis, the odontoblasts form dentin and the dentinal tubules, and their presence within the tubules makes dentin a living responsive tissue.

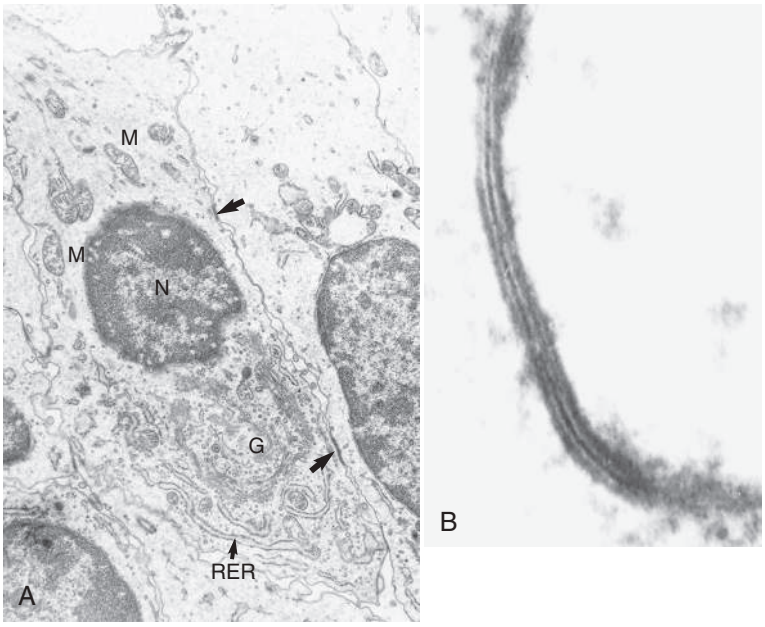


Fig. 13.3 A, Electron micrograph of a mouse molar odontoblast demonstrating gap junctions *(arrows)*, nucleus *(N)*, mitochondria *(M)*, Golgi complex *(G)*, and rough endoplasmic reticulum *(RER)*. **B,** High magnification of a section fixed and stained with lanthanum nitrate to demonstrate a typical gap junction. (Courtesy Dr. Charles F. Cox, School of Dentistry, University of Alabama.)

Dentinogenesis, osteogenesis, and cementogenesis are in many respects quite similar, and odontoblasts, osteoblasts, and cementoblasts have many characteristics in common. Each of these cells produces a matrix composed of collagen fibrils, noncollagenous proteins, and proteoglycans that are capable of undergoing mineralization. The ultrastructural characteristics of odontoblasts, osteoblasts, and cementoblasts are likewise similar in that each exhibits a highly ordered rough endoplasmic reticulum (RER), a prominent Golgi complex, secretory granules, and numerous mitochondria. In addition, these cells are rich in RNA, and their nuclei contain one or more prominent nucleoli. These are the general characteristics of protein-secreting cells.

The most significant differences among odontoblasts, osteoblasts, and cementoblasts are their morphologic characteristics and the anatomic relationship between the cells and the mineralized structures they produce. Whereas osteoblasts and cementoblasts are polygonal to cuboidal in form, the fully developed odontoblast of the coronal pulp is a tall columnar cell.[63,245] In bone and cementum, some of the osteoblasts and cementoblasts become entrapped in the matrix as osteocytes or cementocytes, respectively. The odontoblast, on the other hand, leaves behind a cellular process to form the dentinal tubule, and the cell body resides outside the mineralized tissue. Lateral branches between the major odontoblast processes interconnect[180,258] through canaliculi, just as osteocytes and cementocytes are linked through the canaliculi in bone and cementum. This provides a pathway for intercellular communication and circulation of fluid and metabolites through the mineralized matrix.

The ultrastructural features of the odontoblast have been the subject of numerous investigations. The cell body of the active odontoblast has a large nucleus that may contain up to four nucleoli (Fig. 13.4). The nucleus is situated at the basal end of the cell and is contained within a nuclear envelope. A well-developed Golgi complex, centrally located in the supranuclear cytoplasm, consists of an assembly of smooth-walled vesicles and cisternae. Numerous mitochondria are evenly distributed throughout the cell body. RER is particularly prominent, consisting of closely stacked cisternae forming parallel arrays that are dispersed diffusely within the cytoplasm. Numerous ribosomes closely associated with the membranes of the cisternae mark the sites of protein synthesis. Within the lumen of the cisternae, filamentous material (probably representing newly synthesized protein) can be observed.

The odontoblast appears to synthesize mainly type I collagen,[212,408] although small amounts of type V collagen have been found in the extracellular matrix (ECM). In addition to proteoglycans[39,85,125] and collagen,[212,224] the odontoblast secretes dentin sialoprotein[41] and phosphophoryn,[41,70]

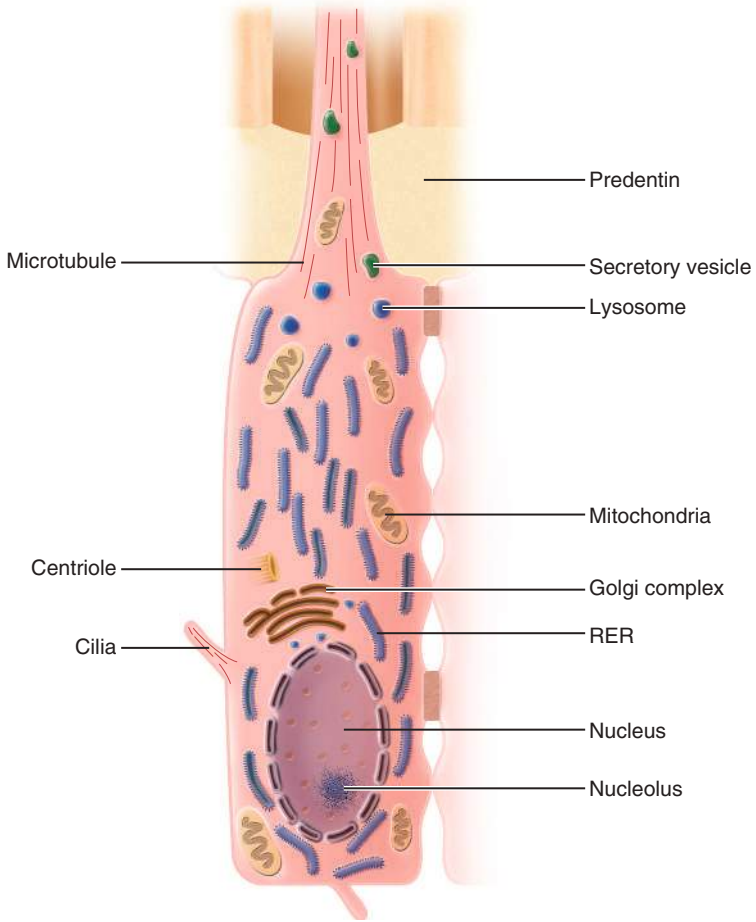


Fig. 13.4 Diagram of a fully differentiated odontoblast. *RER,* Rough endoplasmic reticulum.

a highly phosphorylated phosphoprotein involved in extracellular mineralization.[41,78] Phosphophoryn is unique to dentin and is not found in any other mesenchymal cell types.[78] The odontoblast also secretes both acid phosphatase and alkaline phosphatase. The latter enzyme is closely linked to mineralization, but the precise role of alkaline phosphatase in dentinogenesis is not completely understood. Acid phosphatase, a lysosomal enzyme, may be involved in digesting resorbed material from the predentin matrix.[89]

In contrast to the active odontoblast, the resting or inactive odontoblast has a decreased number of organelles and may become progressively shorter.[63,245] These changes can begin with the completion of root development and eruption when dentin production shifts from primary to secondary dentin.

Direct actions of odontoblasts on dental nerves and vice versa have been proposed based on the excitability of odontoblasts, the differential expression of receptors for neuropeptides on odontoblasts (Fig. 13.5), the demonstration of the thermosensitive transient receptor potential (TRP) ion channels, and the finding that all nine voltage-gated sodium channels are variably expressed on odontoblasts in developing, mature, and aging rat teeth.[53,72,84,109,236,237] In addition, a possible function of odontoblast in immune regulation has been proposed by the finding of innate immune components in the odontoblast layer.[387] Hence the odontoblasts should be capable of recognizing and differentially responding to bacterial components, thereby serving immune and pulp-dentine barrier functions.

ODONTOBLAST PROCESS

A dentinal tubule forms around each of the major odontoblastic processes. The odontoblast process occupies most of the space within the tubule and coordinates the formation of peritubular dentin.

Microtubules and microfilaments are the principal ultrastructural components of the odontoblast process and its lateral branches.[112,160] Microtubules extend from the cell body out into the process.[112] These straight structures follow a course that is parallel with the long axis of the cell and impart the impression of rigidity. Although their precise role is unknown, theories as to their functional significance suggest that they may be involved in cytoplasmic extension, transport of materials, or the provision of a structural framework. Occasionally, mitochondria can be found in the process where it passes through the predentin.

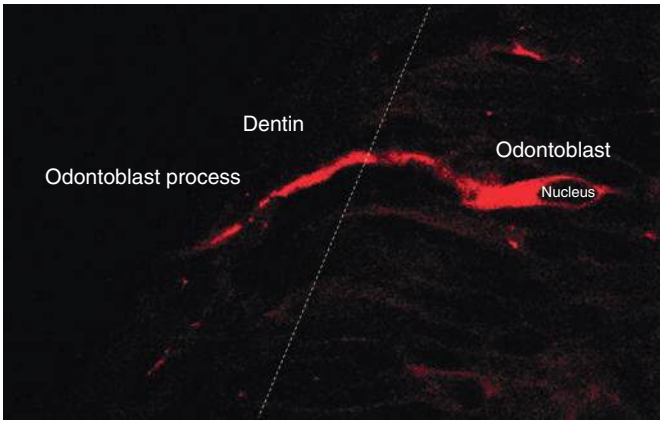


Fig. 13.5 Confocal microscopic image showing an odontoblast with its process expressing neurokinin 2 receptor. Neurokinin 2 has affinity to all the neuropeptides of the neurokinin family. The dotted line represents the border of the predentin. (From Fristad I, Vandevska-Radunovic V, Fjeld K, et al: NK1, NK2, NK3 and CGRP1 receptors identified in rat oral soft tissues, and in bone and dental hard tissue cells, *Cell Tissue Res* 311:383–391, 2003.)

The plasma membrane of the odontoblast process closely approximates the wall of the dentinal tubule. Localized constrictions in the process occasionally produce relatively large spaces between the tubule wall and the process. Such spaces may contain collagen fibrils and fine granular material that presumably represents ground substance (see also The Pulpal Interstitium and Ground Substance in this chapter). The peritubular dentin matrix within the tubule is lined by an electron-dense limiting membrane called the *lamina limitans*.[265,361,405] A narrow space separates the limiting membrane from the plasma membrane of the odontoblast process, except for the areas where the process is constricted.

In restoring a tooth, the removal of enamel and dentin often disrupts odontoblasts.[42,46,74,81,206,262] It would be of considerable clinical importance to establish the extent of the odontoblast processes in human teeth, because with this knowledge the clinician would be in a better position to estimate the impact of the restorative procedure on the underlying odontoblasts. However, the extent to which the process penetrates into dentin has been a matter of considerable controversy. It has long been thought that the process is present throughout the full thickness of dentin. Although ultrastructural studies using transmission electron microscopy have described the process as being limited to the inner third of the dentin,[49,113,361,405] it should be noted that this could possibly be the result of shrinkage occurring during fixation and dehydration. Other studies employing scanning electron microscopy have described the process extending further into the tubule, often as far as the dentoenamel junction (DEJ),[132,181,343,419] but it has been suggested that what has been observed in scanning electron micrographs is actually the lamina limitans.[361,362,405]

In an attempt to resolve this issue, monoclonal antibodies directed against microtubules were used to demonstrate tubulin in the microtubules of the process. Immunoreactivity was observed throughout the dentinal tubule, suggesting that the process extends throughout the thickness of dentin.[343] However, a study employing confocal microscopy found that odontoblast processes in rat molars do not extend to the outer dentin or DEJ, except during the early stages of tooth development.[49] It is likely that the walls of tubules contain many proteins originally derived from odontoblasts that no longer remain at that site. Because dentin matrix has no turnover, these antigens remain fixed in place. From a clinical perspective, it is important to remember that these processes in the tubules represent appendages from living odontoblasts in the pulp, which explains why the dentin must be considered a vital tissue, the destruction of which will affect the pulp.

The odontoblast is considered to be a fixed postmitotic cell in that once it has fully differentiated, it apparently cannot undergo further cell division. If this is indeed the case, the life span of the odontoblast coincides with the life span of the viable pulp. However, its metabolic activity can be dynamically altered (described under the heading Pulpal Repair).

RELATIONSHIP OF ODONTOBLAST STRUCTURE TO SECRETORY FUNCTION

Studies using radiolabeled chemicals have shed light on the functional significance of the cytoplasmic organelles of the active odontoblast.[407,408] In experimental animals, intraperitoneal injection of a collagen precursor (e.g., ^{3}H-proline) was followed by autoradiographic labeling of the odontoblasts and predentin matrix (Fig. 13.6).[408] Rapid incorporation of the isotope in the RER soon lead to labeling of the Golgi complex in the area where the procollagen was packed and concentrated into secretory vesicles. Radiolabeled vesicles passed along their migration pathway until they reach the base of the odontoblast process. Here they fused with the cell membrane and released their tropocollagen molecules into the predentin matrix by the process of exocytosis.

It is now known that collagen fibrils precipitate from a solution of secreted tropocollagen and that the aggregation of fibrils occurs on the outer surface of the odontoblast plasma membrane. Fibrils are released into the predentin and increase in thickness as they approach the mineralized matrix. Whereas fibrils at the base of the odontoblast process are approximately 15 nm in diameter, fibrils in the region of the calcification front have attained a diameter of about 50 nm.

Similar tracer studies[407] have elucidated the pathway of synthesis, transport, and secretion of the predentin proteoglycans. The protein moiety of these molecules is synthesized by the RER of the odontoblast, whereas sulfation and addition of the glycosaminoglycan (GAG) moieties to the protein molecules take place in the Golgi complex. Secretory vesicles then transport the proteoglycans to the base of the odontoblast process, where they are secreted into the predentin matrix. Proteoglycans, principally chondroitin sulfate, accumulate near the calcification front. The role of the proteoglycans is speculative, but mounting evidence suggests that they act as inhibitors of calcification by binding calcium. It appears that just before calcification, the proteoglycans are removed, probably by lysosomal enzymes secreted by the odontoblasts.[86]

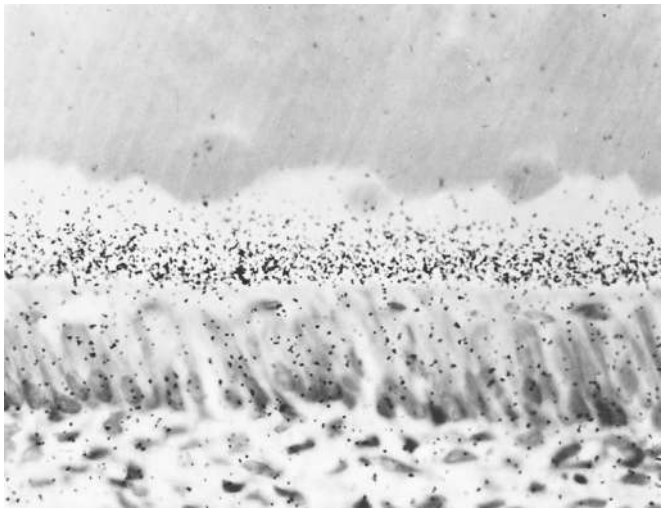


Fig. 13.6 Autoradiograph demonstrating odontoblasts and predentin in a developing rat molar 1 hour after intraperitoneal injection of ^{3}H-proline.

PULP FIBROBLAST

Fibroblasts are the most numerous cells of the pulp. They appear to be tissue-specific cells capable of giving rise to cells that are committed to differentiation (e.g., odontoblast-like cells) if given the proper signal. These cells synthesize type I and III collagen, as well as proteoglycans and GAGs. Thus they produce and maintain the matrix proteins of the ECM. Because they are also able to phagocytose and digest collagen, fibroblasts are responsible for collagen turnover in the pulp.

Although distributed throughout the pulp, fibroblasts are particularly abundant in the cell-rich zone. The early differentiating fibroblasts are polygonal and appear to be widely separated and evenly distributed within the ground substance. Cell-to-cell contacts are established between the multiple processes that extend out from each of the cells. Many of these contacts take the form of gap junctions that provide for electronic coupling or chemical signaling from one cell to another. In terms of ultrastructure, the organelles of the immature fibroblasts are generally in a rudimentary stage of development, with an inconspicuous Golgi complex, numerous free ribosomes, and sparse RER. As they mature, the cells become stellate in form, and the Golgi complex enlarges, the RER proliferates, secretory vesicles appear, and the fibroblasts take on the characteristic appearance of protein-secreting cells. In addition, collagen fibrils accumulate along the outer surface of the cell body. With an increase in the number of blood vessels, nerves, and collagen fibers, there is a relative decrease in the number of fibroblasts in the pulp.

Many fibroblasts of the pulp are characterized by being relatively undifferentiated. A more modern term for these undifferentiated cells is MSC. MSC derived from different tissues appear to be tissue specific and are responsible for maintenance of specific tissue functions. However, MSC share some common characteristics, as they are able to differentiate into other cell lineages when given the proper signals. Like MSC from bone and adipose tissues, pulpal MSC proliferate well in culture (Fig. 13.7), and when stimulated they have neurogenic, adipogenic, chondrogenic, and

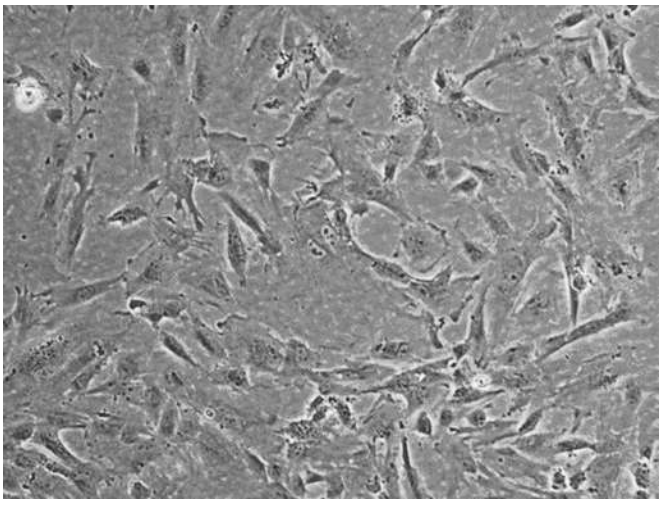


Fig. 13.7 Morphology of cultured pulpal cells. Many of the cells will have mesenchymal stem cell characteristics.

osteogenic differentiation potential.[282] Many pulpal cells seem to remain in a relatively undifferentiated modality, compared with fibroblasts of most other connective tissues.[138] This perception has been supported by the observation of large numbers of reticulin-like fibers in the pulp. Reticulin fibers have an affinity for silver stains and are similar to the argyrophilic fibers of the pulp. However, in a careful review, it appears that actual reticulin fibers may not be present in the pulp; instead the previously described fibers are actually *argyrophilic collagen fibers*.[15] The fibers apparently acquire a GAG sheath, and it is this sheath that is impregnated by silver stains. In the young pulp, the nonargyrophilic collagen fibers are sparse, but they progressively increase in number as the pulp ages.

Many experimental models have been developed to study wound healing in the pulp, particularly dentinal bridge formation after pulp exposure or pulpotomy. One study[100] demonstrated that mitotic activity preceding the differentiation of replacement odontoblasts appears to occur primarily among perivascular fibroblasts.

Pulpal fibroblasts seem to take active part in signaling pathways in the dental pulp. For example, fibroblast growth and synthesis are stimulated by neuropeptides; in turn, fibroblasts produce nerve growth factor (NGF) and proinflammatory cytokines during inflammation.[32,415,420] NGF plays an important role not only in tooth development but also in regulating neuronal and possibly odontoblast responses to injury via activation of similar neurotrophin receptors expressed on both cell types (see also Plasticity of Intradental Nerve Fibers in this chapter).[415]

MACROPHAGE

Macrophages are monocytes that have left the bloodstream, entered the tissues, and differentiated into various subpopulations. The different subpopulations can be studied by their antigenic properties in immunohistochemical studies. Many are found in close proximity to blood vessels. A major subpopulation of macrophages is active in endocytosis and phagocytosis. Because of their mobility and phagocytic activity, they are able to act as scavengers, removing extravasated red blood cells, dead cells, and foreign bodies from the tissue. Ingested material is destroyed by the action of lysosomal enzymes. Another subset of macrophages participates in immune reactions by processing antigen and presenting it to memory T cells.[287] The processed antigen is bound to class II major histocompatibility complex (MHC) molecules on the macrophage, where it can interact with specific receptors present on naive or memory T cells (Fig. 13.8).[135] Such interactions are essential for T-cell–dependent immunity. Similar to fibroblasts, macrophages take an active part in the signaling pathways in the pulp. When activated by the appropriate inflammatory stimuli, macrophages are capable of producing a large variety of soluble factors, including interleukin (IL) 1, tumor necrosis factor (TNF), growth factors, and other cytokines. One study showed that a subset of macrophages express lymphatic markers, indicating a link between macrophages and lymphatic function and development.[19]

DENDRITIC CELL

Dendritic cells are accessory cells of the immune system. Similar cells are found in the epidermis and mucous membranes, where they are called *Langerhans cells*.[175,286] Dendritic cells are primarily found in lymphoid tissues, but they are also widely distributed in connective tissues, including the pulp (Fig. 13.9).[325] These cells are termed *antigen-presenting cells* and are characterized by dendritic cytoplasmic processes and the presence of class II MHC complexes on their cell surface. In the normal pulp they are mostly located in the periphery of the coronal pulp close to the predentin, but they migrate centrally in the pulp after antigenic challenge.[428] They are known to play a central role in the induction of T-cell–dependent immunity. Like antigen-presenting macrophages, dendritic cells engulf protein antigens and then present an assembly of peptide fragments of the antigens and MHC class II molecules. It is this assembly that T cells can recognize. Then the assembly binds to a T-cell receptor and T-cell activation occurs (see Fig. 13.8). Fig. 13.10 shows a cell-to-cell contact between a dendritic-like cell and a lymphocyte.

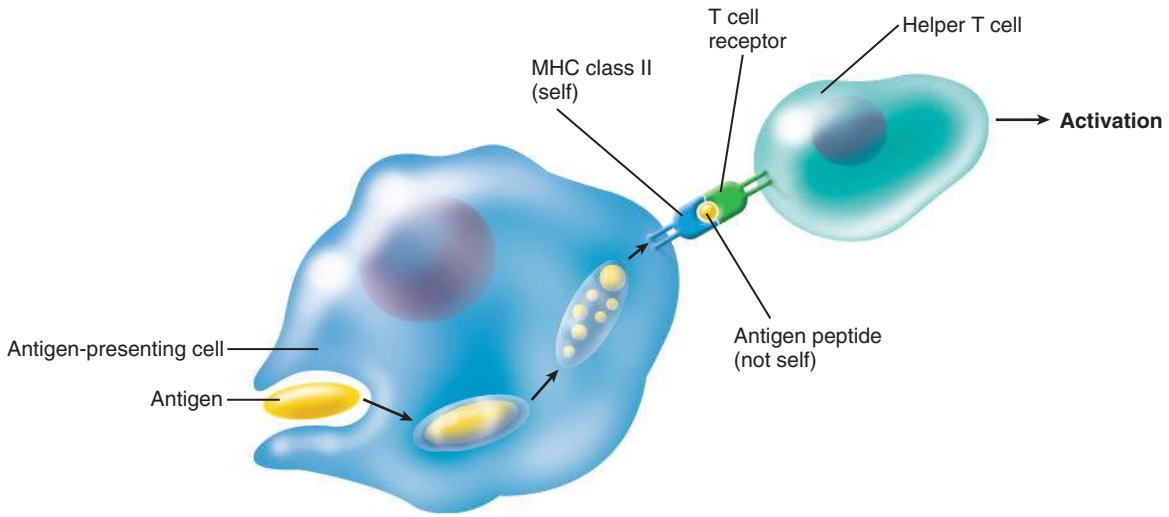


Fig. 13.8 Function of major histocompatibility complex *(MHC)* class II molecule-expressing cells. They act as antigen-presenting cells that are essential for the induction of helper T-cell-dependent immune responses.

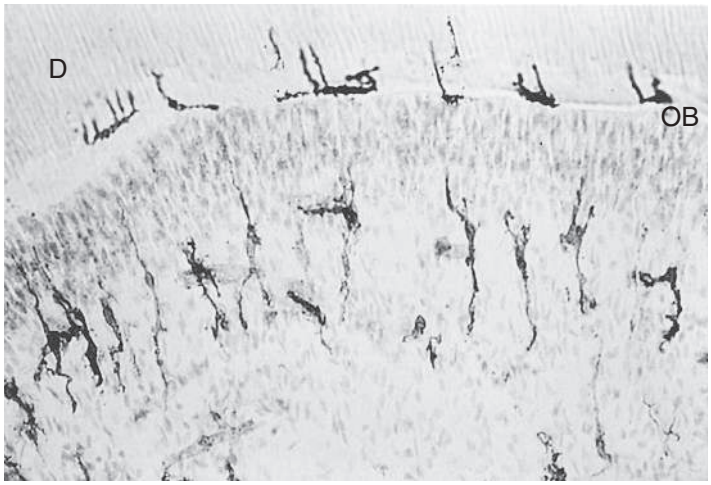


Fig. 13.9 Class II antigen-expressing dendritic cells in the pulp and dentin border zone in normal human pulp, as demonstrated by immunocytochemistry. *D,* Dentin; *OB,* odontoblastic layer.

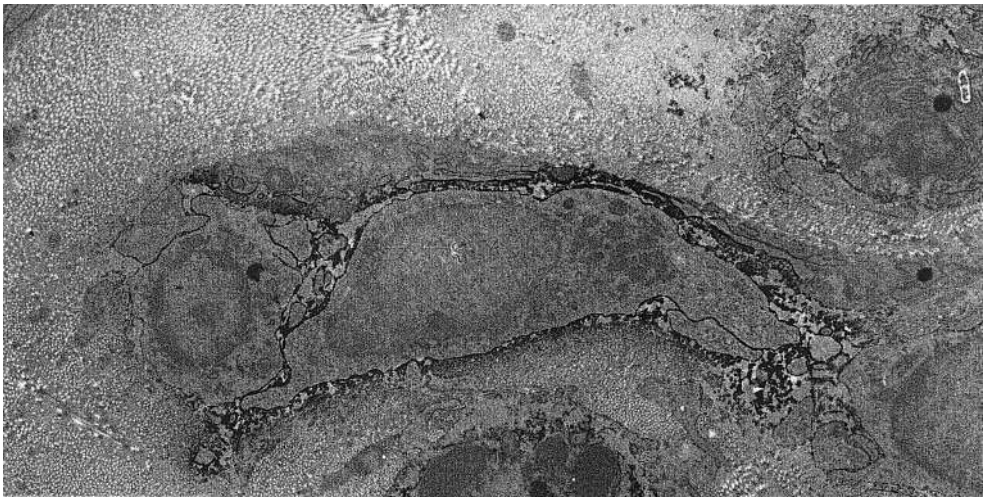


Fig. 13.10 Immunoelectron micrograph of a cell resembling a dendritic cell and a lymphocyte. They show cell-to-cell contact.

LYMPHOCYTE

Hahn and colleagues[135] reported finding T lymphocytes in normal pulps from human teeth. T8 (suppressor) lymphocytes were the predominant T-lymphocyte subset present in these samples. Lymphocytes have also been observed in the pulps of impacted teeth.[205] The presence of macrophages, dendritic cells, and T lymphocytes indicates that the pulp is well equipped with cells required for the initiation of immune responses.[175,325] B lymphocytes are scarcely found in the normal uninflamed pulp.

MAST CELL

Mast cells are widely distributed in connective tissues, where they occur in small groups in relation to blood vessels. Mast cells are seldom found in the normal pulp tissue, although they are routinely found in chronically inflamed pulps.[325] The mast cell has been the subject of considerable attention because of its dramatic role in inflammatory reactions. The granules of mast cells contain heparin, an anticoagulant, and histamine, an important inflammatory mediator, as well as many other chemical factors.

Metabolism

The metabolic activity of the pulp has been studied by measuring the rate of oxygen consumption and the production of carbon dioxide (CO_2) or lactic acid.[26,97,98,99,137,326] An investigation employed an oxygen-sensitive microelectrode inserted into a rat incisor pulp with a micromanipulator, and the authors reported that odontoblasts consumed O_2 at the rate of 3.2 ± 0.2 mL/min/100 g of pulp tissue.[425]

Because of the relatively sparse cellular composition of the pulp, the rate of oxygen consumption is low in comparison with that of most other tissues. During active dentinogenesis, metabolic activity is much higher than after the crown is completed. As would be anticipated, the greatest metabolic activity is found in the region of the odontoblast layer, and the lowest is found in the central pulp, where most of the nerves and blood vessels are located.[25]

In addition to the usual glycolytic pathway, the pulp has the ability to produce energy through a phosphogluconate (i.e., pentose phosphate) shunt type of carbohydrate metabolism,[99] a metabolic pathway that permits tissues to function under varying degrees of ischemia. This could explain how the pulp manages to withstand periods of low perfusion resulting from vasoconstriction induced by epinephrine-containing infiltration anesthesia.[193]

Several commonly used dental materials (e.g., eugenol, zinc oxide and eugenol, calcium hydroxide, silver amalgam) inhibit oxygen consumption by pulp tissue, indicating that these agents may be capable of depressing the metabolic activity of pulpal cells.[98,174] One study[137] found that application of orthodontic force to human premolars for 3 days resulted in a 27% reduction in respiratory activity in the pulp. This study utilized carbon-14-labeled succinic acid in the medium. As cells metabolize succinic acid, they produce $^{14}CO_2$ that can be trapped and quantitated by a liquid scintillation counter.[137] This technique requires only a few milligrams of tissue.

The Pulpal Interstitium and Ground Substance

The interstitium consists of the interstitial fluid and the interstitial (extracellular) matrix and occupies the extracellular and extravascular space. It is amorphous and generally regarded as a gel rather than a solid. Its constituents are similar in all tissues, but their relative amount varies. The major structural component of the interstitium is collagen (Fig. 13.11). The network of collagen fibers also supports the other components of the interstitium, the proteoglycans, hyaluronan, and elastic fibers. The two former components represent the GAGs of the interstitial matrix.

Because of its content of polyanionic polysaccharides, the interstitium is responsible for the water-holding properties of connective tissues and acts as a molecular sieve in regulating the diffusion of substances through this space.[411]

Connective tissue consists of cells and fibers, both embedded in ground substance or ECM. Cells that produce connective tissue fibers also synthesize the major constituents of the ECM. Whereas the fibers and cells have recognizable shapes, the ECM is described as being amorphous. It is generally regarded as a gel rather than a solid. Nearly all proteins of the ECM are glycoproteins.[242] Proteoglycans are an important subclass of glycoproteins.[125] These molecules support cells, provide tissue turgor, and mediate a variety of cell interactions. They have in common the presence of GAG chains and a protein core to which the chains are linked. Except for heparan sulfate and heparin, the chains are composed of disaccharides. The primary function of GAG chains is to act as adhesive molecules that can bond to cell surfaces and other matrix molecules.

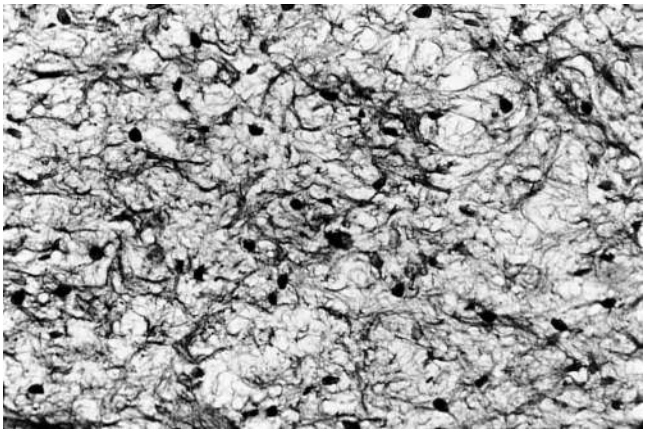


Fig. 13.11 Delicate network of pulpal collagen fibers as demonstrated by the Pearson silver impregnation method.

Fibronectin is a major surface glycoprotein that, together with collagen, forms an integrated fibrillary network that influences adhesion, motility, growth, and differentiation of cells. Laminin, an important component of basement membranes, binds to type IV collagen and cell surface receptors.[120] Tenascin is another substrate adhesion glycoprotein.

In the pulp, the principal proteoglycans include hyaluronic acid dermatan sulfate, heparan sulfate, and chondroitin sulfate.[132] The proteoglycan content of pulp tissue decreases approximately 50% with tooth eruption.[224] During active dentinogenesis, chondroitin sulfate is the principal proteoglycan, particularly in the odontoblast and predentin layer, where it is somehow involved with mineralization; with tooth eruption, hyaluronic acid and dermatan sulfate increase, and chondroitin sulfate decreases greatly.

The consistency of a connective tissue (e.g., the pulp) is largely determined by the proteoglycan components of the ground substance. The long GAG chains of the proteoglycan molecules form relatively rigid coils constituting a network that holds water, thus forming a characteristic gel. Hyaluronic acid in particular has a strong affinity for water and is a major component of ground substance in tissues with a large fluid content, such as Wharton jelly of the umbilical cord. The water content of the young pulp is very high (approximately 90%), so the ground substance forms a cushion capable of protecting cells and vascular components of the tooth.

Ground substance also acts as a molecular sieve in that it excludes large proteins. Cell metabolites, nutrients, and wastes pass through the ground substance between cells and blood vessels. In some ways, ground substance can be likened to an ion exchange resin, because the polyanionic chains of the GAGs bind cations. In addition, osmotic pressures can be altered by excluding osmotically active molecules. Thus proteoglycans can regulate the dispersion of interstitial matrix solutes, colloids, and water, and (in large measure) they determine the physical characteristics of a tissue, such as the pulp.

Degradation of ground substance can occur in certain inflammatory lesions that have a high concentration of macrophage lysosomal enzymes. Proteolytic enzymes, hyaluronidases, and chondroitin sulfatases of lysosomal and bacterial origin are examples of the hydrolytic enzymes that can attack components of the ground substance. The pathways of inflammation and infection are strongly influenced by the state of polymerization of the ground substance components.

HYALURONAN

Another major structural component of the interstitial matrix is hyaluronan. It is an unbranched, random-coil molecule from repeating nonsulfated disaccharide units

and is found in the interstitium as free molecules or bound to cells, possibly via the connection to fibronectin.[209] Its large molecular weight together with its protein structure accounts for its unique properties. It has a high viscosity even at low concentration, exhibits exclusion properties, and has a strong affinity for water.

Hyaluronan is one of several types of GAGs in the pulp.[221,242] The hyaluronan receptor-1 is expressed on lymphatic vessels and also on immune cells in the dental pulp.[19] Hyaluronan is removed from the tissue by the lymphatics and metabolized in the lymph nodes[103] and by endothelial cells in the liver.[136,304]

ELASTIC FIBERS

Elastic fibers constitute an elastin core and a surrounding microfibrillar network and provide elasticity to the tissue.[295] The amount of elastin in the interstitial matrix in most tissue is small. There is no evidence for elastic fibers in the matrix in the pulp.[136,304]

THE INFLAMED INTERSTITIUM

Hyaluronidases, chondroitin sulfatases, and matrix metalloproteinases (MMPs) are examples of the enzymes that can attack components of the interstitium.

During infection and inflammation, the physical properties of the pulp tissue may then be altered due to production of such degrading enzymes.[147,324] In addition to their own damaging effect, they may also pave the way for the deleterious effects of bacterial toxins, increasing the magnitude of the damage.[179]

The pathways of inflammation and infection are strongly influenced by the particular composition of the interstitium in every tissue and its degradation by either host or microbial enzymes.

Connective Tissue Fibers of the Pulp

Two types of structural proteins are found in the pulp: collagen and elastin. Elastin fibers are confined to the walls of arterioles and, unlike collagen, are not a part of the ECM.

A single collagen molecule, referred to as *tropocollagen,* consists of three polypeptide chains, designated as either α-1 or α-2 depending on their amino acid composition and sequence. In the human pulp, the amount of collagen is reported to be 26% to 32% of dry weight in premolars and molars.[383] Type I and type III collagen represent the major subtypes of collagen in the pulp, and type I is found in thick striated fibrils throughout the pulp tissue.[210,342] The different combinations and linkages of chains making up the tropocollagen molecule have allowed collagen fibers and fibrils to be classified into a number of types:

- Type I collagen is found in skin, tendon, bone, dentin, and pulp.
- Type II collagen is found in cartilage.
- Type III collagen is found in most unmineralized connective tissues. It is a fetal form found in the dental papilla and the mature pulp. In bovine pulp, it constitutes 45% of the total pulp collagen during all stages of development.[383]
- Types IV and VII collagen are components of basement membranes.
- Type V collagen is a constituent of interstitial tissues.
- Type VI collagen is a heterotrimer of three distinct chains, α 2 (VI) and α 3 (VI), and is widely distributed in low concentrations in soft tissues at interfibrillar filaments.

Type I collagen is synthesized by odontoblasts and osteoblasts; fibroblasts synthesize types I, III, V, and VII collagen.

In collagen synthesis, the protein portion of the molecule is formed by the polyribosomes of the RER of connective tissue cells. The proline and lysine residues of the polypeptide chains are hydroxylated in the cisternae of the RER, and the chains are assembled into a triple-helix configuration in the smooth endoplasmic reticulum. The product of this assembly is termed *procollagen,* and it has a terminal unit of amino acids known as the *telopeptide of the procollagen molecule*. When these molecules reach the Golgi complex, they are glycosylated and packaged in secretory vesicles. The vesicles are transported to the plasma membrane and secreted by way of exocytosis into the extracellular milieu, thus releasing the procollagen. Here the terminal telopeptide is cleaved by a hydrolytic enzyme, and the tropocollagen molecules begin aggregating to form collagen fibrils. It is believed that the GAGs somehow mediate aggregation of tropocollagen. The conversion of soluble collagen into insoluble fibers occurs as a result of cross-linking of tropocollagen molecules. The transcription factor SOX9 is strongly expressed in normal dental pulp tissue and reduced in inflamed pulp.[232] The factor seems to promote collagen synthesis and seems to be important for ECM balance.

The presence of collagen fibers passing from the dentin matrix between odontoblasts into the dental pulp has been reported in fully erupted teeth.[29] Larger collagen fiber bundles are much more numerous in the radicular pulp than in the coronal pulp. The highest concentration of these larger fiber bundles is usually found near the apex (Fig. 13.12).[371]



Fig. 13.12 Dense bundles of collagen fibers *(CF)* in the apical pulp.

The Trigeminal System

INNERVATION

Pain is a subjective phenomenon involving not only sensory physiologic responses, but also emotional, conceptual, and motivational aspects of behavior. The existence of peripheral "nociceptive" (pain-detecting) sensory neurons forms the basis for pain, and pain sensations of varying qualities and intensities are evoked by activation of the intradental nerves innervating teeth. Noxious stimuli in teeth are transmitted in primary afferent neurons located in the trigeminal ganglion via second-order neurons in the brain stem to the brain (Fig. 13.13) (see also Chapter 4 and later in this chapter). Transmission of sensory information consists of a cascade of events involving input, processing, and sensing,[335] so the control of dental pain should be based on an understanding of the origin of pain signals and the complex modulation that may take place locally and at higher levels. The sensory system of the pulp appears to be well suited for signaling potential damage to the tooth. The tooth is innervated by a large number of myelinated and unmyelinated axons. The number of axons entering a human premolar may reach 2000 or more, and each axon can arborize to form multiple points of innervation.[92,171,172]

Regardless of the nature of the sensory stimulus (i.e., thermal, mechanical, chemical, electric [e.g., pulp tester]), almost all afferent impulses generated from pulp tissue result in the sensation of pain. However, when the pulp is weakly stimulated by an electric pulp tester under carefully controlled experimental conditions, a nonpainful sensation (i.e., prepain) has been reported.[252] Thus not all afferent neurons that innervate the pulp are nociceptors. The innervation of the pulp includes both *afferent neurons,* which conduct sensory impulses, and *autonomic or efferent neurons,*[165] which provide neurogenic modulation of the microcirculation, inflammatory reactions,[150] and perhaps regulate dentinogenesis.[47]

The sympathetic innervation of teeth derives from the superior cervical ganglion (SCG).[8,307] Postganglionic sympathetic nerves travel with the internal carotid nerve, join the trigeminal nerve at the ganglion, and supply teeth and supporting structures via the maxillary and mandibular division of the trigeminal nerve.[244] Sympathetic fibers appear with blood vessels at the time the vascular system is established in the dental papilla.[106] In the adult tooth pulp, sympathetic fibers form plexuses, usually around pulpal arterioles (Fig. 13.14). Stimulation of these fibers results in constriction of the arterioles and a decrease in blood flow.[1,82] The sympathetic neuron terminals contain the classic neurotransmitter, norepinephrine (NE), and neuropeptide Y (NPY) (see Neuropeptides later in this chapter). NPY is synthesized in sympathetic neurons and supplied to terminals by axonal transport. By contrast, NE is mainly produced locally in the terminals. Compared with the sensory nerves, these fibers are most often located in deeper parts of the pulp proper, but fibers have also been found in close relation to odontoblasts.[1,166]

The presence of parasympathetic cholinergic nerves in dental tissues has been and is still controversial, although it has been concluded that there is absence of parasympathetic vasodilation in the cat dental pulp.[292,329] It has been reported that the neuropeptide vasoactive intestinal polypeptide (VIP) is localized in the parasympathetic neurons.[229,230] The origin of VIP-containing fibers in the pulp is uncertain insofar as no form of surgical denervation has resulted in complete loss of these fibers from the dental pulp.[397]

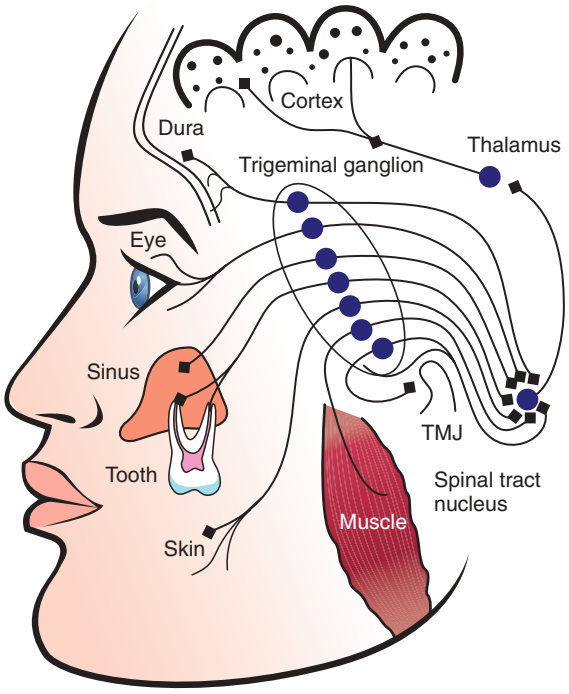


Fig. 13.13 Schematic drawing illustrating the convergence of sensory information from teeth to higher brain centers. *TMJ,* Temporomandibular joint.

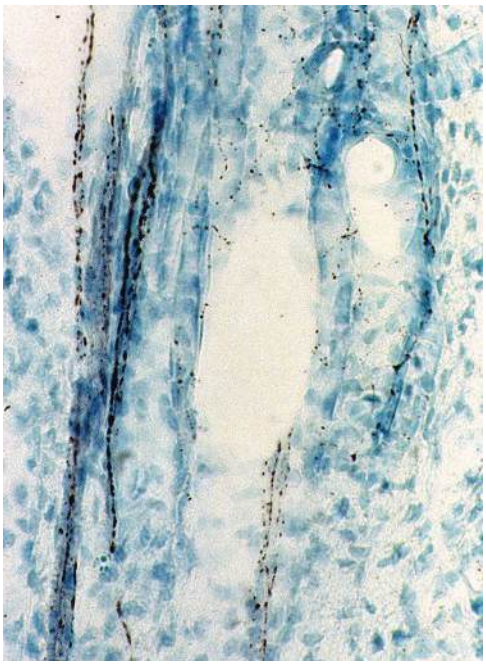


Fig. 13.14 Histologic section, immunohistologically stained for neuropeptide Y (NPY), shows the distribution of sympathetic nerves in the root pulp of a rat molar. NPY fibers are seen associated with blood vessels. (Courtesy Dr. Inge Fristad, Department of Clinical Dentistry, University of Bergen.)

Sensory nerve fibers are usually classified according to their diameter, conduction velocity, and function (Table 13.1). The pulp contains two types of sensory nerve fibers: myelinated (A fibers) and unmyelinated (C fibers). It has been shown that there is some functional overlap between pulpal A and C fibers, as both fiber types can be nociceptors.[165,171,172,251,272] The A fibers include both A-β and A-δ fibers. The A-β fibers may be slightly more sensitive to stimulation than the A-δ fibers, but functionally these fibers are grouped together in the dental pulp, because both innervate the dentinal tubules, and both are stimulated by dentinal fluid movement (Fig. 13.15). Approximately 90% of the A fibers in dental pulp are A-δ fibers.[251] Table 13.2 summarizes the principal characteristics of the main sensory fibers.

During the bell stage of tooth development, "pioneer" nerve fibers enter the dental papilla following the path of blood vessels.[106] Although only unmyelinated fibers are observed in the dental papilla, a proportion of these fibers are probably A fibers that have lost or not developed their myelin sheath. Myelinated fibers are the last major structures to appear in the developing human dental pulp.[11] The number of nerve fibers gradually increases, and branching occurs as the fibers approach dentin. During the bell stage, very few fibers enter the predentin.[106]

The sensory nerves of the pulp arise from the trigeminal nerve and pass into the radicular pulp in bundles by way of the foramen in close association with arterioles and venules (Fig. 13.16). Each of the nerves entering the pulp is invested within Schwann cells, and the A fibers acquire their myelin sheath from these cells. With the completion of root development, the myelinated fibers appear grouped in bundles in the central region of the pulp (Fig. 13.17). Most of the unmyelinated C fibers entering the pulp are located within these fiber bundles; the remainder is situated toward the periphery of the pulp (see Fig. 13.15).[313] It should be noted that single neurons branch and innervate the pulps of multiple teeth in animal studies.[157] Assuming a similar innervation pattern in humans, such organization may contribute to the poor localization of dental pain and may also allow neurogenic vasodilation and inflammatory reactions to

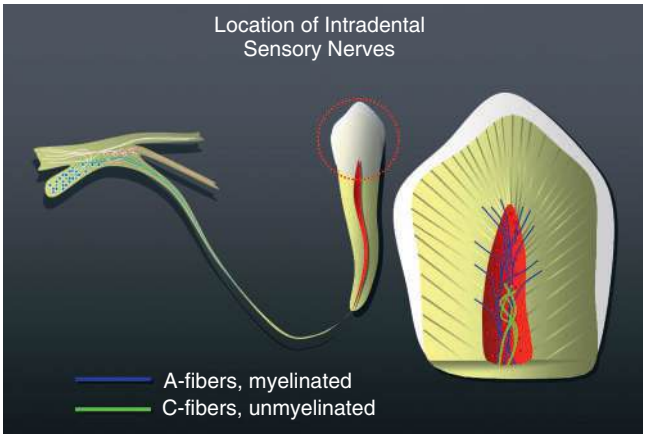


Fig. 13.15 Schematic drawing illustrating the location of A and C fibers in the dental pulp. Myelinated A fibers are located in the periphery of the pulp, penetrating the inner part of dentin. Unmyelinated C fibers are located in the deeper part of the pulp proper.

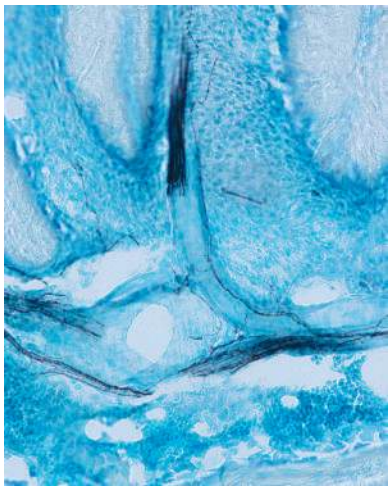


Fig. 13.16 Histologic section, immunohistologically stained for calcitonin gene-related peptide (CGRP), shows distribution of sensory nerves in the apical area of a rat molar. Nerve fibers are seen associated with blood vessels and enter the dental pulp in nerve bundles. (Courtesy Dr. Inge Fristad, Department of Clinical Dentistry, University of Bergen.)

Table 13.1 Classification of Nerve Fibers

Type of Fiber	Function	Diameter (μm)	Conduction Velocity (m/s)
A-α	Motor, proprioception	12–20	70–120
A-β	Pressure, touch	5–12	30–70
A-γ	Motor, to muscle spindles	3–6	15–30
A-δ	Pain, temperature, touch	1–5	6–30
B	Preganglionic autonomic	<3	3–15
C dorsal root	Pain	0.4–1	0.5–2
Sympathetic	Postganglionic sympathetic	0.3–1.3	0.7–2.3

Table 13.2 Characteristics of Sensory Fibers

Fiber	Myelination	Location of Terminals	Pain Characteristics	Stimulation Threshold
A-δ	Yes	Principally in region of pulp-dentin junction	Sharp, pricking	Relatively low
C	No	Probably distributed throughout pulp	Burning, aching, less bearable than A-δ fiber sensations	Relatively high, usually associated with tissue injury

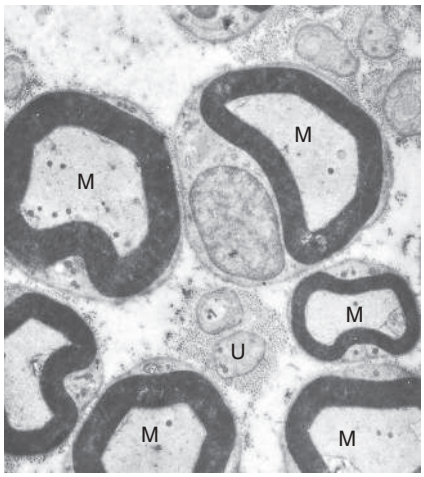


Fig. 13.17 Electron micrograph of the apical pulp of a young canine tooth, showing in cross section myelinated nerve axons *(M)* within Schwann cells. Smaller, unmyelinated axons *(U)* are enclosed singly and in groups by Schwann cells. (Courtesy Dr. David C. Johnsen, School of Dentistry, Case Western Reserve University.)

occur in an area of tissue wider than that affected by the original insult (Fig. 13.18). This phenomenon, also known as the axon reflex,[328] was first demonstrated in rat teeth and later confirmed in cat teeth where stimulation of a second premolar caused neurogenic inflammation in the canine tooth.[157,199] An alternative explanation to this clinical observation is that the pulp has a relatively low density of proprioceptors, and thus patients have difficulty identifying the inflamed tooth until the inflammation reaches the periradicular tissue, which is highly innervated with proprioceptors. This is discussed in greater detail in Chapter 4.

In the human premolar, the number of unmyelinated axons entering the tooth at the apex reached a maximal number shortly after tooth eruption.[172] At this stage, an average of 1800 unmyelinated axons and more than 400 myelinated axons were found, although in some teeth fewer than 100 myelinated axons were present. Five years after eruption, the number of A fibers gradually increased to more than 700. The relatively late appearance of A fibers in the pulp may help to explain why the electric pulp test tends to be unreliable in young teeth, as A fibers are more easily electrically stimulated than C fibers.[110]

A quantitative study of axons 1 to 2 mm coronal to the root apex of fully developed human canine and incisor teeth[172] reported a mean of about 360 myelinated axons in canines and incisors, whereas there were 1600 to 2200 unmyelinated axons. However, this does not reflect the actual number of neurons supporting a single tooth, because multiple branching of the axons may occur in the peripheral tissues. Overall, approximately 80% of the axons were unmyelinated fibers.[171,172]

The nerve bundles pass upward through the radicular pulp together with blood vessels. Once they reach the coronal pulp, they fan out beneath the cell-rich zone, branch into smaller bundles, and finally ramify into a plexus of single-nerve axons known as the *plexus of Raschkow* (Fig. 13.19). Full development of this plexus does not occur until the final stages of root formation.[92] It has been estimated that each axon entering the pulp sends at least eight branches to the plexus of Raschkow. There is prolific branching of the fibers in the plexus, producing a tremendous overlap of receptor fields.[145,272,273,274,278] It is in the plexus that the A fibers emerge from their encircling Schwann cells and branch repeatedly to form the subodontoblastic plexus. Finally, terminal axons pass between the odontoblasts as free nerve endings (see Figs. 13.19–13.21). The extent to which dentin is innervated has been the subject of numerous investigations.[43,46,47,50,92,219] With the exception of the innervation of dentinal tubules, discussed later in this chapter, the bulk of dentin is devoid of sensory nerve fibers. This offers an explanation as to why pain-producing agents (e.g., potassium chloride) do not always elicit pain when applied to exposed dentin. Similarly, application of topical anesthetic solutions to dentin does not decrease its sensitivity. Hence a high concentration of lidocaine solution is needed to block the response of intradental nerves to mechanical stimulation of the dentin.[4]

One investigator[133] studied the distribution and organization of nerve fibers in the dentin-pulp border zone of human teeth. On the basis of their location and pattern of branching, several types of nerve endings were described (Fig. 13.22). Some fibers were found running from the subodontoblastic nerve plexus toward the odontoblast layer. However, these fibers do not reach the predentin; they terminate in extracellular spaces in the cell-rich zone, the cell-poor zone, or the odontoblast layer. Other fibers extend into the predentin and run through a dentinal tubule in close association with an odontoblast process. Most of these intratubular fibers extend into the dentinal tubules for only a few micrometers, but a few may penetrate as far as 100 μm (see Figs. 13.19 and 13.20). The area covered by a single such terminal complex often reaches thousands of square micrometers.[133,273]

Intratubular nerve endings are most numerous in the area of the pulp horns, where as many as 40% of the tubules may contain fibers.[47,219] The number of intratubular fibers decreases in other parts of the dentin, and in root dentin only about 1% of dentinal tubules contain fibers.

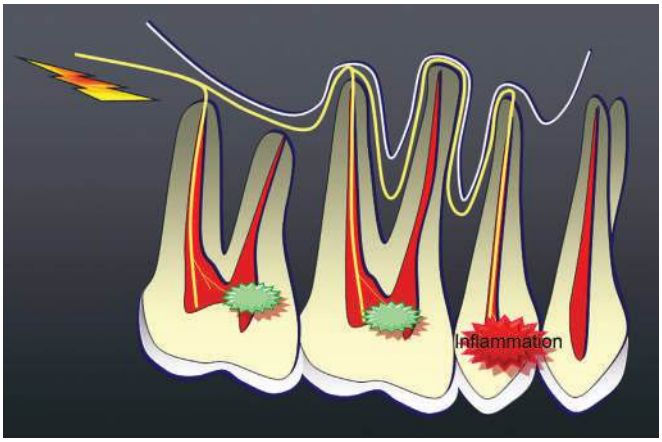


Fig. 13.18 Branches of a single nerve fiber may innervate more than a single tooth. Nerve fiber activation in one tooth may thus result in neurogenic inflammation in adjacent teeth innervated by the same axon. The mechanism is known as the axonal reflex. This phenomenon may also contribute to the poor localization of pain reported by patients.

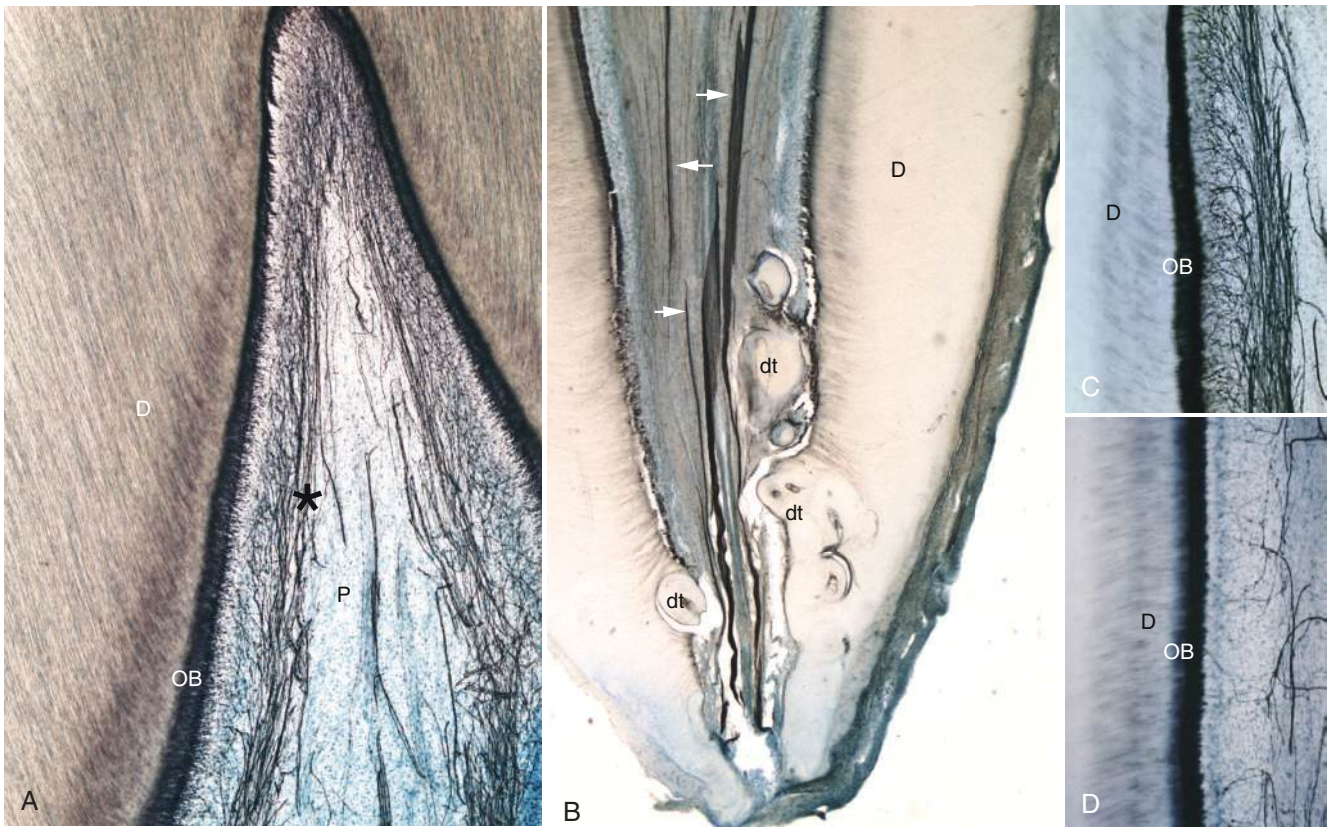


Fig. 13.19 Histological sections from a human premolar labeled with the neurochemical marker protein gene product (PGP) 9.5, showing the dental pulp and surrounding dentin. **A,** Nerve fibers entering the coronal pulp *(P)* form a dense network beneath the odontoblasts *(OB)*, known as the plexus of Raschkow *(*)*. **B,** Nerves are entering the apical area and pass through the radicular pulp as nerve bundles *(arrows)*. Attached and free denticles *(dt)* are seen close to dentin in the apical area. **C** and **D,** Compared to dense peripheral branching of nerves in the coronal pulp **(C)**, the branching of nerves in the radicular pulp is sparse **(D)**. (Courtesy Dr. Inger Hals Kvinnsland.)

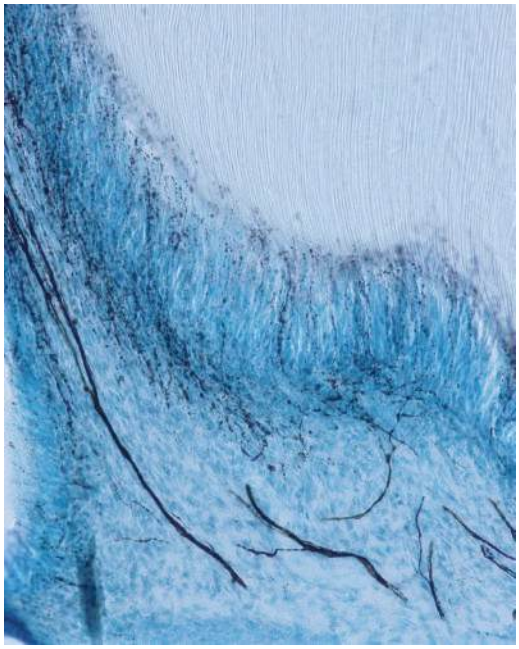


Fig. 13.20 Histologic section, immunohistologically stained for calcitonin gene-related peptide (CGRP), shows the distribution of sensory nerves in a rat molar. Nerves enter the coronal pulp in bundles and ramify in a network beneath the odontoblasts (i.e., plexus of Raschkow), before entering between the odontoblasts and the inner part of dentin.

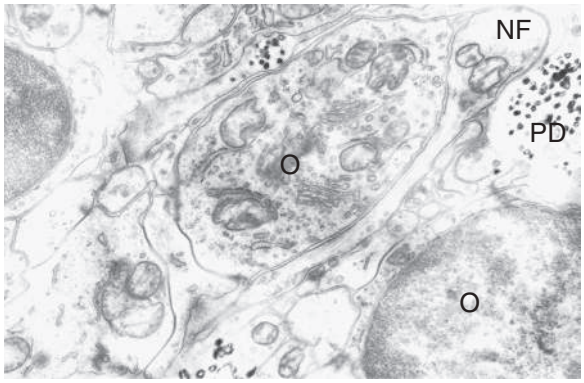


Fig. 13.21 Unmyelinated nerve fiber *(NF)* without a Schwann cell covering located between adjacent odontoblasts *(O)* overlying pulp horn of a mouse molar tooth. Predentin *(PD)* can be seen at upper right. Within the nerve, there are longitudinally oriented fine neurofilaments, microvesicles, and mitochondria. (From Corpron RE, Avery JK: The ultrastructure of intradental nerves in developing mouse molars, *Anat Rec* 175:585, 1973.)

This notion has been challenged in a study that stained pulps for protein gene-product 9.5, a specific marker for nerves.[235] In that study, root dentin appeared to be as well innervated as coronal dentin. The anatomic relationship between the odontoblast processes and sensory nerve endings has led to much speculation as to the functional relationships between these structures, if any.[46] When present,

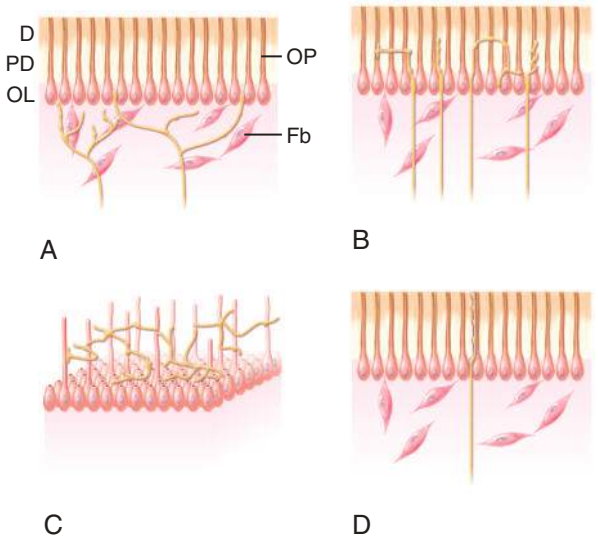


Fig. 13.22 Schematic drawing showing distribution of nerve fibers in the dentin-pulp border zone. **A,** Fibers running from the subodontoblastic plexus to the odontoblast layer. **B,** Fibers extending into the dentinal tubules in the predentin. **C,** Complex fibers that branch extensively in the predentin. **D,** Intratubular fibers extending into the dentin. *D,* Dentin; *Fb,* fibroblast; *OL,* odontoblast layer; *OP,* odontoblast process; *PD,* predentin.

nerve fibers lie in a groove or gutter along the surface of the odontoblast process, and toward their terminal ends they twist around the process like a corkscrew. The cell membranes of the odontoblast process and the nerve fiber are closely approximated and run closely parallel for the length of their proximity, but they are not synaptically linked.[162]

Although it may be tempting to speculate that the odontoblasts and their associated nerve axons are functionally interrelated and that together they play a role in dentin sensitivity, there is a paucity of evidence supporting this hypothesis. If the odontoblast were acting as a classic receptor cell,[†] it would have chemical, electric, or mechanical communication with the adjacent nerve fiber. However, researchers have been unable to find classic anatomic structures (e.g., synaptic junctions) that could functionally couple odontoblasts and nerve fibers together. With regard to the membrane properties of odontoblasts, it has been reported that the membrane potential of the odontoblast is low (−24 to −30 mV),[202,231] and that the cell does not respond to electric stimulation.[292,413] It would appear that the odontoblast does not possess the properties of an excitable cell. Further, the sensitivity of dentin is not diminished after disruption of the odontoblast layer.[38,220] It is still possible that odontoblasts could modulate neuronal function via alteration in sodium channel activity or the release of paracrine factors that diffuse to the closely approximated nerve terminal.

Another study showed that a reduction in pulpal blood flow (PBF), induced by stimulation of sympathetic fibers leading to the pulp, results in depressed excitability of pulpal A fibers.[83] The excitability of C fibers is less affected than that of A fibers by a reduction in blood flow.[370]

Of clinical interest is the evidence that nerve fibers of the pulp may be resistant to necrosis[83,260] because their cell bodies are found in ganglia outside the pulp. Because nerve bundles in general are more resistant to autolysis than other tissue elements, even in degenerating pulps, C fibers might still be able to respond to noxious stimulation. It may be that C fibers remain excitable even after blood flow has been compromised in the diseased pulp, as C fibers are often able to function in the presence of hypoxia.[370] This may explain why instrumentation of the root canals of apparently nonvital teeth sometimes elicits pain. On the other hand, histologic studies on nonvital teeth failed to demonstrate high levels of innervation, leading to the suggestion that pain may be due to the transfer of noxious chemicals to terminals located in periapical tissues.[260]

STEPS AND MECHANISMS IN PAIN PERCEPTION

When activated by a stimulus sufficient to cause tissue damage or release of inflammatory mediators, nerve endings in the pulp and periradicular tissues begin to send bursts of messages to the central nervous system (CNS) that may eventually be perceived as pain. The anatomic pathway for this transmission of information has been fairly well established, and it is tempting to view the perception of pain of orofacial origin as a simple graded response to the intensity of the stimulus. However, researchers have come to realize that the pain system is a complex, multilevel system that begins with the *detection* of tissue-damaging stimuli in the periphery, the *processing* of that input at the level of the medullary spinal cord, and the *perception* of what is felt as pain in higher brain regions such as the cerebral cortex (Fig. 13.23). After a noxious stimulus is detected in the periphery, there is ample opportunity for a great deal of endogenous and possibly exogenous modification of the message prior to its ultimate perception. The clinician deals with all three levels of the pain system in diagnosing and treating odontalgia, and a practitioner with a basic understanding of each level will be able to recognize therapeutic opportunities and apply effective pain control methods.

DETECTION: THE FIRST STEP IN PAIN PERCEPTION

Various types of peripheral neurons are found in the trigeminal system, including large-diameter, heavily myelinated Aα, Aβ, and Aγ fibers associated with motor, proprioception, touch, pressure, and muscle spindle stretch functions. But it is the smaller, less myelinated Aδ and yet smaller and unmyelinated C fibers that conduct information likely to be perceived as pain. These two classes of pain-sensing nerve fibers, or *nociceptors,* are both found in the tooth pulp, but there are three to eight times more unmyelinated C fibers than A-δ fibers.[42,46,172,376] It should be noted that this classification system is based purely on the size and myelination of the neurons and does not necessarily indicate function. For example, another class of pulpal C fibers are the postganglionic sympathetic efferents found in association with blood vessels, where

[†]A receptor cell is a non–nerve cell capable of exciting adjacent afferent nerve fibers. Synaptic junctions connect receptor cells to afferent nerves.

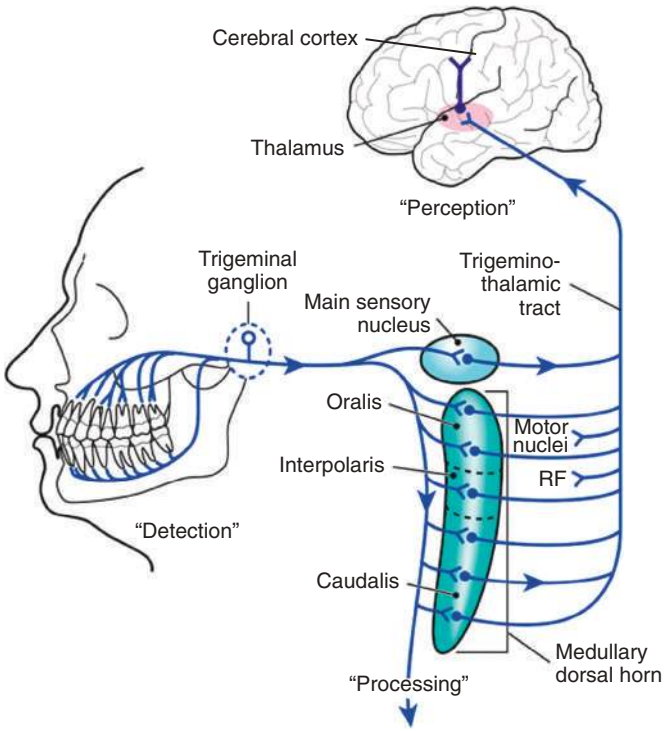


Fig. 13.23 Schematic diagram of the pathway of transmission of nociceptive information from the orofacial region. The trigeminal pain system is a complex multilevel system that begins with the *detection* of tissue damaging stimuli in the periphery, the *processing* of that input at the level of the medullary spinal cord, and the final *perception* of what is felt as pain in the cerebral cortex. There is growing appreciation for the concept that once a noxious stimulus is detected in the periphery, there is the opportunity for a great deal of modification of the message prior to ultimate perception. *RF,* Reticular formation.

they regulate PBF[2,190,305,306] and may also influence the activity of peripheral nociceptors (for reviews, see Perl[302] or Hargreaves[139]). Because most pulpal sensory fibers are nociceptive, their terminal branches are free nerve endings, and physiologic stimulation by any modality (temperature, hyperosmotic fluids) results in the perception of pure pain, which can be difficult for patients to localize. Under experimental conditions, electrical stimulation can result in a *prepain* sensation that is also difficult to localize. Once inflammation has extended to the periodontal ligament, which is well endowed with Aβ discriminative touch receptors, the localization of pain is more predictable with light mechanical stimuli such as the percussion test.

In the normal uninflamed pulp and periradicular tissues, a noxious stimulus causes depolarization of nociceptors sufficient to generate action potentials by means of the opening of voltage-gated sodium channels (Na_v). After generation of an action potential, not only is information sent to the CNS, but also in an *antidromic* fashion (i.e., in the reverse direction of the impulse, also termed as an axon reflex). In this way, proinflammatory neuropeptides such as substance P (SP), calcitonin gene-related peptide (CGRP), neurokinins, and the classic neurotransmitter, glutamate, are released from afferent terminals in the pulp and periradicular tissues, even including neighboring teeth (see Fig. 13.18).

Neuropeptides

Of immense importance in pulp biology is the presence of neuropeptides in pulpal nerves.[43,47,50,356] Pulpal nerve fibers contain neuropeptides such as CGRP (see Figs. 13.16 and 13.20),[42] SP,[273,291] NPY (see Fig. 13.14), neurokinin A (NKA),[12] and VIP.[233,396] In rat molars, the largest group of intradental sensory fibers contains CGRP. Some of these fibers also contain other peptides, such as SP and NKA.[43,46] Release of these peptides can be triggered by numerous stimuli, including tissue injury,[12] complement activation, antigen-antibody reactions,[286] or antidromic stimulation of the inferior alveolar nerve.[288,291] Once released, vasoactive peptides produce vascular changes that are similar to those evoked by histamine and bradykinin (BK) (i.e., vasodilation).[288] In addition to their neurovascular properties, SP and CGRP contribute to inflammation and promote wound healing.[32,374] The release of CGRP can be modified by sympathetic agonists and antagonists,[117,140] offering the promise of using such agonists to treat dental pain. This latter point is important because clinicians use sympathetic agonists every day—the vasoconstrictors present in local anesthetic solutions may have direct effects on inhibiting dental nerve activity. Local anesthetic reduction in pain may be due to the actions of both the local anesthetic and the vasoconstrictor.[139,140] In cats, capsaicin acutely activates and chronically blocks the *transient receptor potential subtype V1* (TRPV1)-expressing classes of C and A-δ nociceptors in the pulp.[165] In addition, the chronic application of capsaicin ointment to skin has been shown to relieve pain in patients, suggesting that clinical trials evaluating chronic application of capsaicin for treating pulpal or periradicular pain may be of value.

It has been reported[251,291] that mechanical stimulation of dentin produces vasodilation within the pulp, presumably by causing the release of neuropeptides from intradental sensory fibers (neurogenic inflammation).[291] Electric stimulation of the tooth has a similar effect.[154] The pulpal concentrations of CGRP, SP, and NKA are elevated in painful human teeth over healthy control teeth extracted for orthodontic reasons.[12] These peptides are also elevated in pulps beneath advancing carious lesions.[316,317]

PULP TESTING

The electric pulp tester delivers a current sufficient to overcome the resistance of enamel and dentin and stimulate the sensory A fibers at the dentin-pulp border zone. Smaller C fibers of the pulp do not respond to the conventional pulp tester because significantly more current is needed to stimulate them.[272] Bender and associates[17] found that in anterior teeth, the optimal placement site of the electrode is the incisal edge, as the response threshold is lowest at that location and increases as the electrode is moved toward the cervical region of the tooth.

Cold tests using CO_2 snow or liquid refrigerants and heat tests employing heated gutta-percha or hot water activate hydrodynamic forces within the dentinal tubules, which in turn excite the intradental A fibers. C fibers are generally not activated by these tests unless they produce injury to the pulp. It has been shown that cold tests do not injure the pulp.[110] Heat tests have a greater potential to produce injury, but if the tests are used properly, injury is not likely.

SENSITIVITY OF DENTIN

The mechanisms underlying dentin sensitivity have been a subject of interest for many years. How are stimuli relayed from the peripheral dentin to the sensory terminals located in the region of the dentin-pulp border zone? Converging evidence indicates that movement of fluid in the dentinal tubules is the basic event in the arousal of dentinal pain.[36,38,268,392,394,395] It now appears that pain-producing stimuli, such as heat, cold, air blasts, and probing with the tip of an explorer, have in common the ability to displace fluid in the tubules.[36,251] This is referred to as the *hydrodynamic mechanism of dentin sensitivity*. The hydrodynamic theory suggests that dentinal pain associated with stimulation of a sensitive tooth ultimately involves mechanotransduction. Classical mechanotransducers have been recognized on pulpal afferents, providing a mechanistic support to this theory.[151] Thus fluid movement in the dentinal tubules is translated into electric signals by receptors located in the axon terminals innervating dentinal tubules. Using single-fiber recording techniques, a positive correlation was found between the degree of pressure change and the number of nerve impulses leaving the pulp (Figs. 13.24 and 13.25).[251,271,391] Thus the outward fluid movements (negative pressure) produce a much stronger nerve response than inward movements.[251,394]

In experiments on humans, brief application of heat or cold to the outer surface of premolar teeth evoked a painful response before the heat or cold could have produced temperature changes capable of activating sensory receptors in the underlying pulp.[271,377] The evoked pain was of short duration: 1 or 2 seconds. The thermal diffusivity of dentin is relatively low, yet the response of the tooth to thermal stimulation is rapid, often less than 1 second. Evidence suggests that thermal stimulation of the tooth results in a rapid movement of fluid into the dentinal tubules. This results in activation of the sensory nerve terminal in the underlying pulp. Presumably, heat expands the fluid within the tubules faster than it expands dentin, causing the fluid to flow toward the pulp, whereas cold causes the fluid to contract more rapidly than dentin, producing an outward flow. It is speculated that the rapid movement of fluid across the cell membrane of the axon terminal activates a mechanosensitive receptor, similar to how fluid movement activates hair cells in the cochlea of the ear. All axon terminals have membrane channels through which charged ions pass, and this initial receptor current, if sufficient, can trigger voltage-gated sodium channels to depolarize the cell, leading to a barrage of impulses to the brain. Some ion channels are activated by voltage, some by chemicals, and some by mechanical pressure.[251,268,273] In the case of pulpal nerve fibers

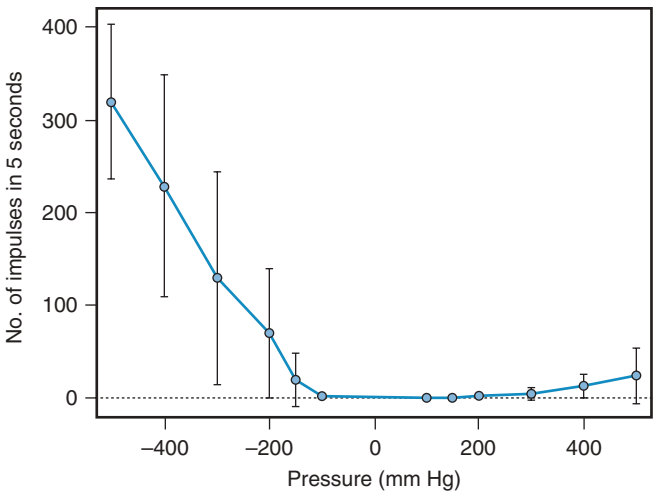


Fig. 13.25 The average number of impulses recorded from dentine after application of pressure stimuli to the dentine. More impulses are recorded after application of negative pressure (outward fluid flow) than after positive pressure (inward fluid flow). (From Vongsavan N, Matthews B: The relationship between the discharge of intradental nerves and the rate of fluid flow through dentine in the cat, *Arch Oral Biol* 52:643, 2007.)

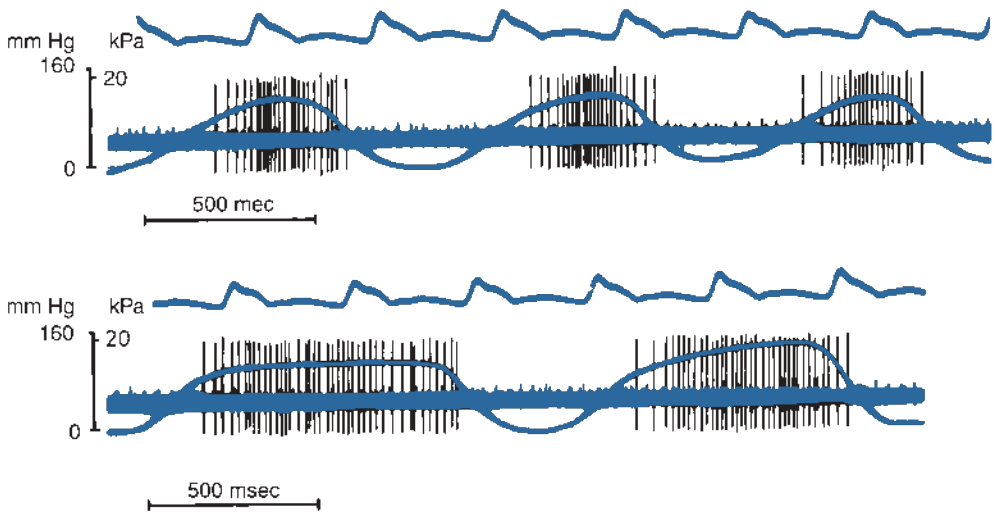


Fig. 13.24 Response of a single dog pulp nerve fiber to repeated hydrostatic pressure stimulation pulses. Lower solid wavy line of each recording indicates the stimulation pressure applied to the pulp. Upper line *(kPa)* is the femoral artery blood pressure curve recorded to indicate the relative changes in the pulse pressure and the heart cycle. (Modified from Närhi M: Activation of dental pulp nerves of the cat and the dog with hydrostatic pressure, *Proc Finn Dent Soc* 74[suppl 5]:1, 1978.)

that are activated by hydrodynamic forces, pressure would be transduced, gating mechanosensitive ion channels.

The dentinal tubule is a capillary tube with an exceedingly small diameter.[‡,113] The physical properties of capillarity are significant because fluid force increases as the diameter is reduced. If fluid is removed from the outer end of exposed dentinal tubules by dehydrating the dentinal surface with an air blast or absorbent paper, capillary forces produce a rapid outward movement of fluid in the tubule (Fig. 13.26). According to Brännström,[36] desiccation of dentin can theoretically cause dentinal fluid to flow outward at a rate of 2 to 3 mm/s. In addition to air blasts, dehydrating solutions containing hyperosmotic concentrations of sucrose or calcium chloride can produce pain if applied to exposed dentin.

Investigators have shown that it is the A fibers rather than the C fibers that are activated by hydrodynamic stimuli (e.g., heat, cold, air blasts) applied to exposed dentin.[270,278] However, if heat is applied long enough to increase the temperature of the dentin-pulp border by several degrees Celsius, then C fibers may respond, particularly if the heat produces injury. It seems that the A fibers are mainly activated by a rapid displacement of the tubular contents.[268] Slow heating of the tooth produced no response until the temperature reached 111°F (43.8°C), at which time C fibers were activated, presumably because of heat-induced injury to the pulp. These C fibers are called *polymodal nociceptors* because they contain numerous receptors that confer the ability to detect and respond to many different types of stimuli.[273,274] Capsaicin, the pungent active ingredient in hot peppers, is known to simulate C fibers and a subset of A-δ fibers.[165] Capsaicin activates a receptor termed the "TRP, subtype vanilloid 1" or TRPV1.[59] The TRPV1 receptor is expressed primarily on a major subclass of nociceptors and responds to heat (>110°F [>43°C]), certain inflammatory mediators, and acid (pH < 6). Thus TRPV1 has been considered a molecular integrator of polymodal noxious stimuli.[283] The ability of the TRPV1 antagonist, capsazepine, to inhibit acid-, heat-, and capsaicin-activated trigeminal neurons[225] has led to the development of new drugs (i.e., TRPV1 antagonists) for the treatment of pulpal pain. Eugenol is known to activate and ultimately desensitize TRPV1, and this may explain the anodyne action of zinc oxide eugenol temporary restorations.[422]

It has also been shown that pain-producing stimuli are more readily transmitted from the dentin surface when the exposed tubule apertures are open[159,173] and the fluid within the tubules is free to flow outward.[173,251,394] For example, acid treatment of exposed dentin to remove the smear layer opens the tubule orifices and makes the dentin much more responsive to stimuli such as air blasts and probing.[159,274]

Perhaps the most difficult phenomenon to explain is dentinal pain associated with light probing of dentin. Even light pressure of an explorer tip can produce strong forces.[§] These forces have been shown to mechanically compress dentin and close open tubule orifices with a smear layer that causes sufficient displacement of fluid to excite the sensory receptors in the underlying pulp (Fig. 13.27).[55,56]

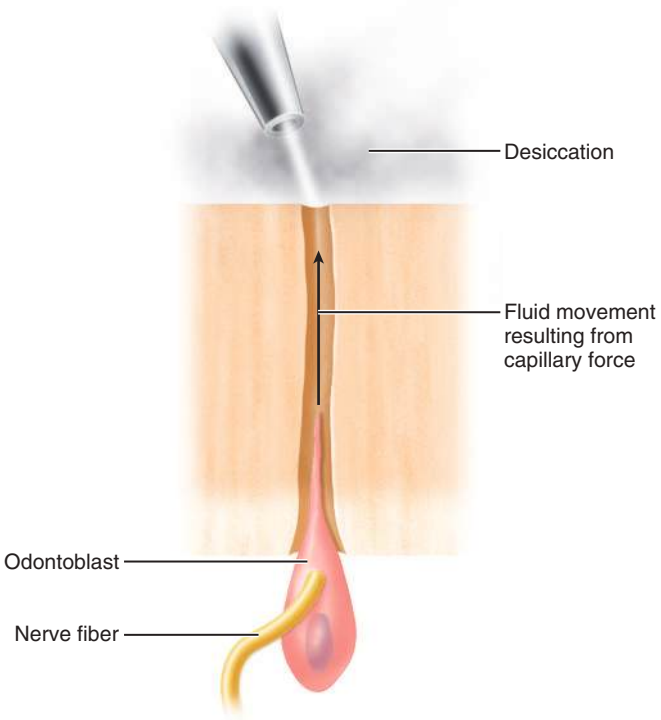


Fig. 13.26 Diagram illustrating movement of fluid in the dentinal tubules resulting from the dehydrating effect of a blast of air from an air syringe.

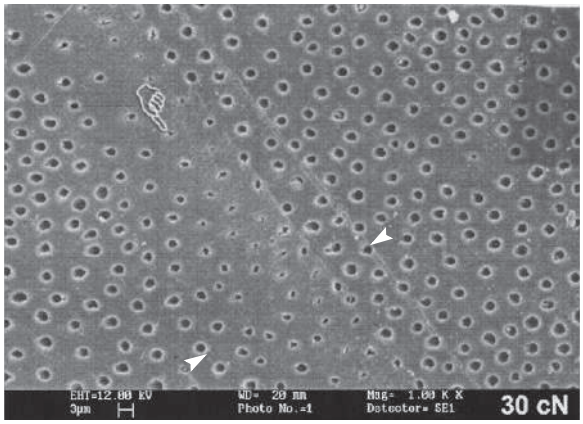


Fig. 13.27 Scanning electron micrograph of a shallow groove *(between white arrowheads)* created in polished dentin by a dental explorer tine under a force of 30 g (30 cN). Note the partial occlusion of the tubules by smeared matrix. (From Camps J, Salomon JP, Meerbeek BV, et al: Dentin deformation after scratching with clinically relevant forces, *Arch Oral Biol* 48:527, 2003.)

[‡]To appreciate fully the dimensions of dentin tubules, understand that the diameter of the tubules (about 1 μm) is much smaller than that of red blood cells (about 7 μm). The thickness of coronal dentin is about 3 mm, so each tubule is 3000 μm long but only 1 μm in diameter. Thus each tubule is 3000 diameters long.

[§]A force of 44 cM (44 g) applied to an explorer having a tip 40 μm in diameter would produce a pressure of 2437 MPa on the dentin.[55] This is far in excess of the compressive strength of dentin, listed as 245 MPa, as evidenced by the shallow grooves lined by smear layers created in dentin using this force.[56]

Considering the density of the tubules in which hydrodynamic forces would be generated by probing, multiple nerve endings would be simultaneously stimulated when a dental explorer is scratched across dentin. Another newly suggested explanation is that the nerves innervating teeth are special in nature and that low-threshold mechanoreceptors are signaling nociceptive input in teeth.[105] Similar low-threshold mechanoreceptors convey tactile sensation in skin. The authors used the term low-threshold "algoneurons" for these nerves. This theory is not in conflict with the hydrodynamic theory but may help to explain the sensation of pain felt after weak mechanical stimuli like air puffs and water spray.

Another example of the effect of strong hydraulic forces that are created within the dentinal tubules is the phenomenon of odontoblast displacement. In this reaction, the nuclei and cell bodies of odontoblasts are displaced upward in the dentinal tubules, presumably by a rapid movement of fluid in the tubules produced when exposed dentin is desiccated, as with the use of an air syringe or cavity-drying agents (Fig. 13.28).[206] Such cellular displacement results in the destruction of odontoblasts, because cells thus affected soon undergo autolysis and disappear from the tubules. Displaced odontoblasts may eventually be replaced by stem cells that migrate from the cell-rich zone of the pulp, as discussed later in the chapter.

The hydrodynamic theory can also be applied to an understanding of the mechanism responsible for hypersensitive dentin.[37,38] There is controversy regarding whether exposed dentin is simply sensitive or becomes truly hypersensitive.[273,274,278] Growing evidence indicates that new sodium channels responsible for activating nerves are expressed in nerve tissue exposed to inflammation.[121,139,314] An increase in the density of sodium channels or their sensitivity may contribute to dentinal hypersensitivity. Hypersensitive dentin is also associated with the exposure of dentin normally covered by cementum or enamel. The thin layer of cementum is frequently lost as gingival recession exposes cementum to the oral environment. Cementum is subsequently worn away by brushing, flossing, or using toothpicks. Once exposed, the dentin may respond to the same stimuli to which any exposed dentin surface responds (e.g., mechanical pressure, dehydrating agents). Although the dentin may at first be very sensitive, within a few weeks the sensitivity usually subsides. This desensitization is thought to occur as a result of gradual occlusion of the tubules by mineral deposits, thus reducing hydrodynamic forces. In addition, deposition of reparative dentin over the pulpal ends of the exposed tubules probably also reduces sensitivity, because reparative dentin is less innervated by sensory nerve fibers.[45] However, some hypersensitive dentin does not spontaneously desensitize, so hypersensitivity may be caused by either inflammatory changes in the pulp or mechanical changes in the patency of dentinal tubules.

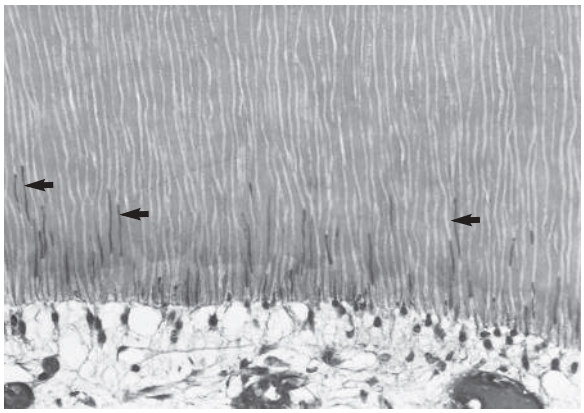


Fig. 13.28 Odontoblasts *(arrows)* displaced upward into the dentinal tubules.

Currently, the treatment of hypersensitive teeth is directed toward reducing the functional diameter of the dentinal tubules to limit fluid movement. Four possible treatment modalities[293,378] can accomplish this objective:

1. Formation of a smear layer on the sensitive dentin by burnishing the exposed root surface[159,273,274]
2. Application of agents, such as oxalate compounds, that form insoluble precipitates within the tubules[159,298]
3. Application of agents such as hydroxyethyl methacrylate (HEMA) with or without glutaraldehyde that are thought to occlude tubules with precipitated plasma proteins in dentinal fluid[331]
4. Application of dentin bonding agents to seal off the tubules[309]

Dentin sensitivity can be modified by laser irradiation, but clinicians must be concerned about its effect on the pulp.[353,363]

PERIPHERAL SENSITIZATION

Following repeated noxious stimuli, both A and polymodal C fiber nociceptors undergo a process of sensitization manifested by three obvious changes in response patterns. First, firing thresholds may decrease, so that previously nonnoxious stimuli may trigger discharges, contributing to the sensation of pain *(allodynia)*. Second, after-discharges may occur, so that noxious stimuli may produce an even greater increase in the perceived intensity of pain *(hyperalgesia)*. And third, firing may occur spontaneously, contributing to the development of spontaneous pain. These changes are often seen in endodontic pain patients and may be explained in part by the effects of chemical mediators released into inflamed pulp and periradicular tissues. Such mediators include substances produced from damaged tissues, agents of vascular origin, and peptides released from the nerve fibers themselves (Table 13.3). Other mechanisms of peripheral sensitization are listed in Box 13.1.

HYPERALGESIA AND ALLODYNIA

Three characteristics of hyperalgesia are (1) spontaneous pain, (2) a decreased pain threshold, and (3) an increased response to a painful stimulus.[139] The peripheral mechanisms for these symptoms include a decrease in firing threshold, an increase in responsiveness to noxious stimuli, and development of spontaneous discharges of nociceptors. All three of these characteristics can be seen in patients experiencing inflammatory pain of pulpal origin (Table 13.4). It is recognized that hyperalgesia can be produced by sustained inflammation, as in the case of sunburned skin. Clinical observation has shown that the sensitivity of dentin often increases when the underlying pulp

Table 13.3 Effect of Inflammatory Mediators on Nociceptive Afferent Fibers

Mediator	Effect on Nociceptors	Effect on Human Volunteers
Potassium[176]	Activate	++
Protons[222,350]	Activate	++
Serotonin[16,176]	Activate	++
Bradykinin[16,176,215]	Activate	+++
Histamine[176]	Activate	+
Tumor necrosis factor-α	Activate	?
Prostaglandins[27]	Sensitize	±
Leukotrienes[27,234]	Sensitize	±
Nerve growth factor[217,303]	Sensitize	++
Substance P[134]	Sensitize	±
Interleukin 1[101]	Sensitize (?)	?

+, Positive; ++, very positive; +++, extremely positive; ±, equivalent; ?, unknown.
Modified from Fields H: *Pain*, New York, 1987, McGraw-Hill.

Box 13.1 Peripheral Mechanisms Contributing to Hyperalgesia and Allodynia

- Composition and concentration of inflammatory mediators[144]
- Changes in afferent fiber: activation and sensitization[203,248,330]
- Changes in afferent fiber: sprouting[52]
- Changes in afferent fiber: proteins[52,121,416]
- Tissue pressure[269,270]
- Tissue temperature[255]
- Sympathetic primary afferent fiber interactions[168,214,301]
- Aβ fiber plasticity[277]

Modified from Hargreaves KM, Swift JQ, Roszkowski MT, et al: Pharmacology of peripheral neuropeptide and inflammatory mediator release, *Oral Surg Oral Med Oral Pathol* 78:503, 1994.

Table 13.4 Signs of Hyperalgesia and Allodynia and Endodontic Diagnostic Tests

Signs of Hyperalgesia	Related Diagnostic Tests or Symptoms
Spontaneous pain	Spontaneous pain
Reduced pain threshold	Percussion test, palpation test, throbbing pain
Increased response to painful stimuli	Increased response to pulp test (electric or thermal test)

From Hargreaves KM, Swift JQ, Roszkowski MT, et al: Pharmacology of peripheral neuropeptide and inflammatory mediator release, *Oral Surg Oral Med Oral Pathol* 78:503, 1994.

becomes acutely inflamed, and the tooth may be more difficult to anesthetize. This is due in part to the upregulation of tetrodotoxin-resistant (TTX-resistant) sodium channels in inflamed neural tissue.[121,139] NGF also seems to play an important role in hyperalgesia.[303] NGF regulates chronic inflammatory hyperalgesia by controlling gene expression in sensory neurons,[279] including genes involved in inflammatory hyperalgesia in the dental pulp.[75] Although a precise explanation for hyperalgesia is lacking, apparently localized elevations in tissue pressure and inflammatory mediators that accompany acute inflammation play an important role.[153,156,351,367,384] Clinically, we know that when the pulp chamber of a painful tooth with an abscessed pulp is opened, drainage of exudate soon produces a reduction in the level of pain. This suggests that mechanical stimuli may contribute substantially to pain during inflammatory hyperalgesia.

From a clinical point of view, *thermal allodynia* is the term that best describes a patient whose chief complaint is, "I have pain when I drink cold beverages." *Mechanical allodynia* is involved when the chief complaint is "It now hurts when I bite on this tooth." These previously nonnoxious stimuli now cause the perception of pain. Hyperalgesia is manifested in endodontic pain patients when noxious stimuli (e.g., refrigerant sprays or CO_2 snow used in the cold test) produce much more pain than they would in teeth with normal pulp tissues. Spontaneous pain involves episodes of pain that seem to be unprovoked. All these changes can be partly explained by sensitization of peripheral nerve endings in the pulp and periradicular tissues.

Many silent nerve fibers are present in the normal pulp[273,274] and are termed *silent* because they are not excited by ordinary external stimuli. Once they are sensitized through pulpal inflammation, they begin to respond to hydrodynamic stimuli.[47,273,274,278] This phenomenon may provide an additional mechanism for dentin hypersensitivity. The molecular mechanisms of this activation are not known in detail but involve the upregulation of numerous genes and their products.[12,46,121]

INFLAMMATORY MEDIATORS

Among the best characterized of the inflammatory mediators are the prostaglandins (PGs), which are derived from arachidonic acid via the action of the cyclooxygenase (COX) enzyme systems. The human COX enzyme is known to exist in at least two forms, COX-1 and COX-2. COX-1 is constitutively expressed and produces PGs that are involved in basic housekeeping functions such as cytoprotection in the stomach, regulation of blood flow in the kidneys, and the formation of thromboxane A_2. The formation of thromboxane A_2 can ultimately lead to platelet aggregation; therefore inhibition of thromboxane A_2 should decrease platelet aggregation. COX-2 is inducible, synthesized in inflamed tissues (including dental pulp),[267] and is important in the production of the proinflammatory PGs as well as the vasodilating prostacyclin (PGI_2). Although they do not produce pain if applied alone, PGs are known to sensitize peripheral nociceptors, which increases the algogenic (pain-producing) properties of serotonin and BK.[77,126] The exact mechanism by which PGs increase neuronal excitability is not clear, but there is a growing body of evidence to suggest that they activate the PG E receptor subtypes EP2 and EP3 in the trigeminal system[299] and exert their effects by regulating the activity of certain ion channels,[401] including voltage-gated sodium channels (for a review, see England[88]). For example, application of prostaglandin E_2 (PGE_2) to isolated dorsal

root ganglion neuronal somata more than doubles the responsiveness of certain sodium channels found predominantly on nociceptors—channels thought to be relatively resistant to lidocaine.[9,122] When administered to rats before an inflammatory insult, ibuprofen, a nonselective COX inhibitor, has been shown to block the increased expression of $Na_v1.7$ and $Na_v1.8$.[129,130] Therefore if the concentrations of PGs in inflamed pulp and periradicular tissues can be decreased with nonsteroidal antiinflammatory drugs (NSAIDs) or corticosteroids, postoperative pain may be relieved; in addition, more profound local anesthesia may be achieved in patients with hyperalgesia of pulpal origin.[142,163,259] It is interesting to note that sensory neurons themselves are a source of PGs; during inflammation, the levels of PGE_2 appear to increase in dorsal root ganglia and spinal cord, suggesting that NSAIDs also have a central site of action (discussed in Chapter 6).[240,412]

BK is a proinflammatory mediator derived from circulating plasma proteins and also causes direct activation of nociceptive neurons, resulting in pain. Increased levels of BK have been demonstrated in the inflamed dental pulp,[211] and the presence of growth factors associated with inflammation (e.g., NGF) have been reported to cause an increase in the expression of mRNA encoding B1 and B2 receptors in primary cultures of rat trigeminal ganglia,[368] as well as other receptors such as TRPV1 and TRPA1.[75,170] The TRPV1 is the "capsaicin receptor"; it plays a key role in mediating inflammatory pain. TRPA1 is expressed on capsaicin-sensitive neurons[75] and interacts with TRPV1.[320] BK likely increases the excitability of nociceptive neurons through its action on TRPV1 and TRPA1 (for review, see Tominaga et al.[364]).

Cytokines are a diverse group of regulatory proteins synthesized and secreted by a variety of cell types, such as leukocytes, neurons, and glia. In particular, TNF-α and the interleukins IL-1β, IL-6, and IL-8 are thought to play a role in the neuroplastic changes that occur in nociceptors innervating inflamed tissues, leading to hyperalgesia.[201] Application of TNF-α rapidly sensitizes TRPV1,[185] contributing to activation of the capsaicin-sensitive class of nociceptors. All of these are thought to exist in the inflamed pulp (for review, see Fouad[102]) and are thought to act at least in part by causing increased release of prostanoids.[354]

PAINFUL PULPITIS

From the foregoing, it is apparent that pain associated with the stimulation of the A fibers does not necessarily signify that the pulp is inflamed or that tissue injury has occurred. Clinically, pain produced by A fibers in response to the hydrodynamic mechanism has a sharp or bright quality as contrasted with the dull, boring, or throbbing pain associated with C fibers. A fibers have a relatively low threshold of excitability to external stimuli,[251,272] and painful pulpitis is more likely to be associated with nociceptive C fiber activity indicative of pulpal tissue injury.[272–274,278] The clinician should carefully examine symptomatic teeth to rule out the possibility of hypersensitive dentin, cracked or leaky fillings, or fracture lines—each of which may initiate hydrodynamic forces—before establishing a diagnosis of reversible or irreversible pulpitis (see also Chapters 1 and 22).

Pain associated with an inflamed or degenerating pulp may be either provoked or spontaneous. The hyperalgesic pulp may demonstrate a lowered threshold of pain by responding to stimuli that usually do not evoke pain (allodynia), or the pain may be exaggerated and persist longer than normal (hyperalgesia).[6] On the other hand, the tooth may commence to ache spontaneously in the absence of any external stimulus.[139] Spontaneous, unprovoked pain generally indicates a pulp that is seriously damaged and generally will not respond to noninvasive therapy.

PLASTICITY OF INTRADENTAL NERVE FIBERS

It has become apparent that the innervation of the tooth is a dynamic process in which the number, size, and cytochemistry of nerve fibers can change because of aging,[104,227,356] tooth injury,[43,46,47,50] and dental caries.[316] For example, in rats, nerve fibers sprout into inflamed tissues surrounding sites of pulpal injury, and the content of CGRP and SP increases in these sprouting fibers.[43,46,47,50,54] When inflammation subsides, the number of sprouts decreases. Fig. 13.29 compares the normal distribution of CGRP-immunoreactive sensory fibers in an adult rat molar with those beneath a shallow cavity preparation. The innervational pattern in normal and inflamed teeth is governed by neuronal growth factors. Neurotrophic and target-derived factors regulate neuronal structure, survival, and function, and are important for the maintenance of neuronal phenotype

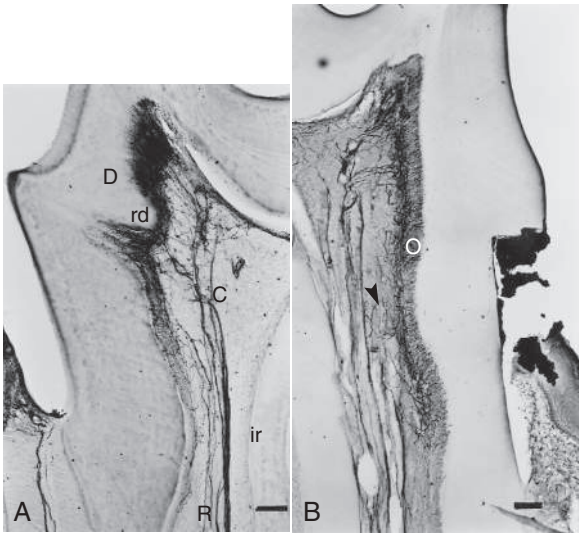


Fig. 13.29 A, Normal distribution of calcitonin gene–related peptide (CGRP)-immunoreactive sensory fibers in adult rat molar. Nerve fibers typically are unbranched in the root *(R)*, avoid interradicular dentin *(ir)*, and form many branches in coronal pulp *(C)* and dentin *(D)*. Nerve distribution is often asymmetric, with endings concentrated near the most columnar odontoblasts (in this case on the left side of the crown). When reparative dentin *(rd)* forms, it alters conditions so that dentinal innervation is reduced. **B,** Shallow class I cavity preparation on the cervical root of a rat molar was made 4 days earlier. Primary odontoblast *(O)* layer survived, and many new CGRP-immunoreactive terminal branches spread beneath and into the injured pulp and dentin. Terminal arbor can be seen branching *(arrowhead)* from a larger axon and growing into the injury site. Scale bar: 0.1 mm. (From Taylor PE, Byers MR and Redd PE: Sprouting of CGRP nerve fibers in response to dentin injury in rat molars, *Brain Res* 461:371–376, 1988.)

characteristics. During development, all dental fibers appear to require NGF and express its receptor, tropomyosin receptor kinase A (TrkA), , at some stages,[249] whereas in adult teeth the large trigeminal neurons are potentially dependent only on dental pulp–derived, glial cell line–derived neurotrophic factor (GDNF); the smaller trigeminal neurons remain dependent on NGF.[204,311] This suggests that GDNF may function as a neurotrophic factor for the subset of larger neurons supporting the tooth, which apparently mediate mechanosensitive stimuli, whereas NGF is suggested to support neurons responsible for nociception.[264] NGF is the most extensively investigated among the trophic factors.[213] Binding of target-derived NGF is dependent on specific TrkA receptors located on the axonal surface, with subsequent internalization and transport to the cell body, where the effects are mediated.

Regulation of neural changes during inflammation seems to be a function of NGF expression.[48,54] NGF receptors are found on intradental sensory fibers and Schwann cells.[46] Evidence indicates that NGF is synthesized by fibroblasts in the coronal subodontoblastic zone (i.e., cell-rich zone), particularly in the tip of the pulp horn.[46] Maximal sprouting of CGRP- and SP-containing nerve fibers corresponds to areas of the pulp with increased production of NGF.[54] Fig. 13.30 shows the expression of NGF-mRNA in a pulp horn subjacent to cavity preparation.

It has been suggested that neuroimmune interactions take place in the dental pulp, because a coordinated increase of pulpal nerves and immune cells has been demonstrated.[175,325] In addition, recruitment of immunocompetent cells has been demonstrated in the dental pulp after electrical tooth stimulation.[67,108] Similar responses have been seen in periapical bone in rats with radicular pulpitis.[198] Neurogenic inflammation is generally thought to enhance healing, because denervated teeth show poorer healing following pulp exposures than innervated teeth.[43,51]

Another consideration in the neural response to inflammation is the possibility of a change in the distribution and activity of voltage-gated sodium channels. In particular, mice lacking the gene for $Na_v1.7$ show reduced painlike behaviors when treated with a variety of proinflammatory agents.[275] Also implicated in the altered firing characteristics of nociceptors innervating inflamed tissues are the sodium channels that are resistant to TTX, the biotoxin found in the tetraodon pufferfish. The two main TTX-R sodium channels are $Na_v1.8$ and $Na_v1.9$, and both have been shown to be increased twofold to fourfold in inflamed dental pulp collected from patients with a diagnosis of irreversible pulpitis.[403,409] When exposed to PGE_2, neurons isolated from dorsal root ganglia cells have been shown to increase TTX-resistant sodium channel currents within minutes,[122] indicating an increased activation of existing channels, rather than de novo protein synthesis. These sodium channels are relatively resistant to lidocaine,[319] and this may explain the difficulty in achieving profound anesthesia in inflamed tissues (see also Chapter 6).[142]

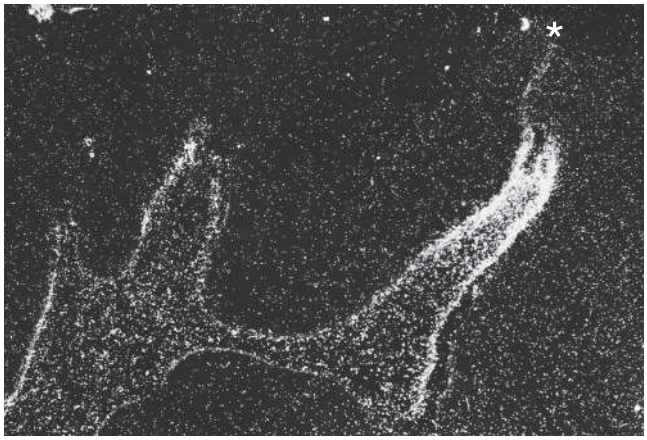


Fig. 13.30 NGF-mRNA is upregulated in the mesial pulp horn 6 hours after cavity preparation. (From Byers MR, Wheeler EF, and Bothwell M: Altered expression of NGF and p75 NGF-Receptor mRNA by fibroblasts of injured teeth precedes sensory nerve sprouting, *Growth Factors* 6:41–45, 1992.)

TISSUE INJURY AND DEAFFERENTATION

When a peripheral nerve is cut or crushed, an interruption of the afferent input to the CNS occurs, which is called *deafferentation.* It would be logical to assume that the result of deafferentation would be anesthesia of the formerly innervated area, but occasionally other symptoms may occur, which may surprisingly include pain. Following nerve injury, a dramatic shift in transcription of neuropeptides, receptors, and sodium channels has been documented. The bidirectional contact between the nerve cell and the peripheral target tissue is lost, and the neurons change into a state of either regeneration or neuronal cell death. The impact on neurons in the trigeminal ganglion is dependent on the injury site. A peripheral injury has less effect than a more centrally located one. However, even a small pulp exposure induces neuronal changes, both in the trigeminal ganglion and at the second-order neuronal level in the brain stem.[44,398] Because each single-rooted tooth contains about 2000 nerve fibers,[171,172] extirpation of the pulp is shown to cause both neurochemical and degenerative changes of their cell bodies in the trigeminal ganglion.[131,188,340] The central projection of these nerves to the spinal nucleus of the trigeminal nerve is also affected,[372] and there is evidence for transsynaptic changes[336,340] that are reflected in the sensory cortex. Even larger responses would be expected from tooth extraction, where both periodontal ligament and pulpal innervation are destroyed.

When an axon is severed peripherally, a complete degeneration of the cell bodies may not always occur.[399] Attempted regeneration by axonal sprouting may result in altered expression of various receptors, resulting in sensitivity to NE (via increased adrenergic receptor activity)[284] or acetylcholine (ACh) (via increased cholinergic receptor activity),[73] sensitizing sensory neurons to autonomic activity. In addition, dorsal horn neurons, deprived of their normal sensory input, may begin to respond to other nearby afferents. Thus normal inhibitory influences are reduced, and a widening of the sensory receptive field is produced, which can produce central sensitization (see Central Sensitization later in this chapter). *Phantom tooth pain* is another term often used synonymously with pain following deafferentation. Different reports suggest an incidence of persistent pain following pulpectomy to be in the range of 3% to 6%.[243,389]

Following tissue injury or tissue inflammation, extensive changes occur in the gene expression of sensory ganglion neurons and, by way of transsynaptic mechanisms, in their central projections (Fig. 13.31).[46,107,188,340] One example is the upregulation of inducible gene transcription factors such as c-*fos*[44] and different subsets of sodium channels.[119] This is thought to result in alterations in threshold properties and the size of receptor fields. C-*fos* is not normally expressed in neurons in the brain stem, but chronic pulpitis causes prolonged increases in c-*fos* expression is some brain stem neurons (see Fig. 13.31).[44]

If such changes also occur in humans, they may help to explain why certain patients may complain of vague, poorly described pain for months following endodontic treatment. If their pulpitis caused sprouting of periapical nerves,[50,51,54,198] then these nerves may have taken part in transport of peripheral signaling molecules to the cell body by way of retrograde axoplasmic flow.[43,54] This could induce changes in the expression of many genes, resulting in central sensitization[6,42] that may require many months to correct.[75] Reports of nerve sprouting in human inflamed pulps have been confirmed in different studies.[316,318,325] Such reactions might contribute to increases in dentin sensitivity as well as expansion of receptive fields.[273,274,278] Sprouting of sympathetic fibers has also been reported,[149] but the timing appears to be different. The functional implications and how these relate to pain mechanisms are unknown, but it has been suggested that these reactions are involved in healing and nociception following pulpal inflammation.[117,149,150]

PROCESSING: THE SECOND STEP IN PAIN PERCEPTION

The Medullary Dorsal Horn

After activation of peripheral nociceptors, nerve impulses in the form of action potentials convey information about the intensity (encoded by firing frequency), quality (encoded by type of neuron activated), and temporal features (encoded by onset, duration, and offset of depolarization) of the peripheral stimuli to the CNS. In the trigeminal pain system, these action potentials arrive at the trigeminal spinal tract nuclear complex located in the medulla.[143,218,329] Three distinct subnuclei can be found in this complex.

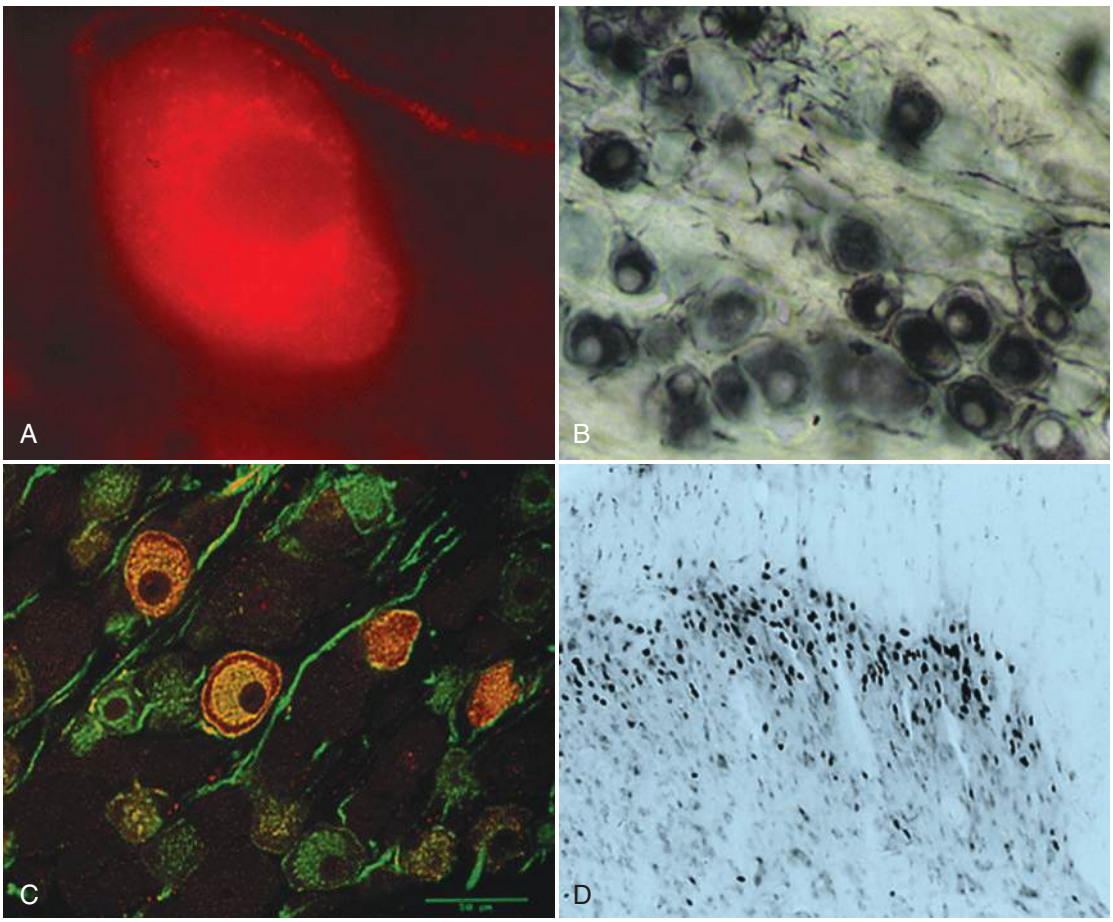


Fig. 13.31 Plastic changes in the trigeminal system. After injury, trigeminal neurons upregulate production of autonomic neuropeptides like neuropeptide Y (NPY) (**A**) and growth-associated protein 43 (GAP 43) connected to neurite outgrowth and regeneration (**B**). **C,** Confocal image showing that NPY is upregulated in injured neurons expressing GAP 43. **D,** The interstitial cells of the trigeminal tract of medullary dorsal horn express c-*fos* during chronic tooth pulp inflammation in rat. (**B** and **C,** Adapted from Khullar SM, Fristad I, Brodin P, et al: Upregulation of growth associated protein 43 expression and neuronal co-expression with neuropeptide Y following inferior alveolar nerve axotomy in the rat, *J Peripher Nerv Sys* 3:79, 1998. **D,** Adapted from Byers MR, Chudler EH, Ladarola MJ: Chronic tooth pulp inflammation causes transient and persistent expression of Fos in dynorphin-rich regions of rat brainstem, *Brain Res* 861:191, 2000.)

Named for their anatomic position, they are the *subnuclei oralis, interpolaris,* and *caudalis* (see Fig. 13.23). Although the more rostral subnuclei (oralis and interpolaris) receive some nociceptive input from oral tissues,[71] most such input is received at the level of the subnucleus caudalis.[238,335] Because of its organizational similarity to the dorsal horn of the spinal cord (which receives nociceptive input from the somatosensory system), the subnucleus caudalis has been termed the *medullary dorsal horn*.

Components of the Medullary Dorsal Horn

The medullary dorsal horn relays information to higher centers in the brain and serves as the site of much potential processing of the signals from primary afferent sensory nerve fibers. Output from this region can be increased (hyperalgesia), decreased (analgesia), or misinterpreted (referred pain). Understanding the functional components involved in such processing not only helps explain some of these clinical phenomena but also allows evaluation of potential therapeutic modalities currently under investigation. Functional components include the central terminals of primary nociceptors (Aδ and C fiber afferents), the second-order projecting neurons, interneurons, the terminals of descending neurons, and glial[246] cells (for review, see Hargreaves[139]).

Primary afferent fibers (whose cell bodies are located in the trigeminal ganglion) transmit signals to projection neurons via the release of transmitters such as the excitatory amino acid, glutamate, and the neuropeptide, SP. Receptors for these neurotransmitters are found on postsynaptic membranes and include the N-methyl-D-aspartate receptor (NMDA) and α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor (AMPA) classes of glutamate receptors and the neurokinin 1 (NK1) class of SP receptors. Antagonists to these receptors have been shown to reduce hyperalgesia in animal studies.[60] In a human clinical trial using an oral surgery model, the AMPA/kainate antagonist, LY293558, was shown to be antihyperalgesic.[119] NK1 antagonists have shown promising results in animal studies, but in general, they have displayed limited analgesic efficacy in humans.[158]

The cell bodies of the second-order (projection) neurons in the trigeminal pain system are found in the medullary dorsal horn; their processes cross the midline and project rostrally to the thalamus via the trigeminothalamic tract (Fig. 13.32). From the thalamus, third-order neurons relay information to the cerebral cortex via a thalamocortical tract. Once signals have reached the cortex, the input may be perceived as pain. Evidence exists that referred pain is

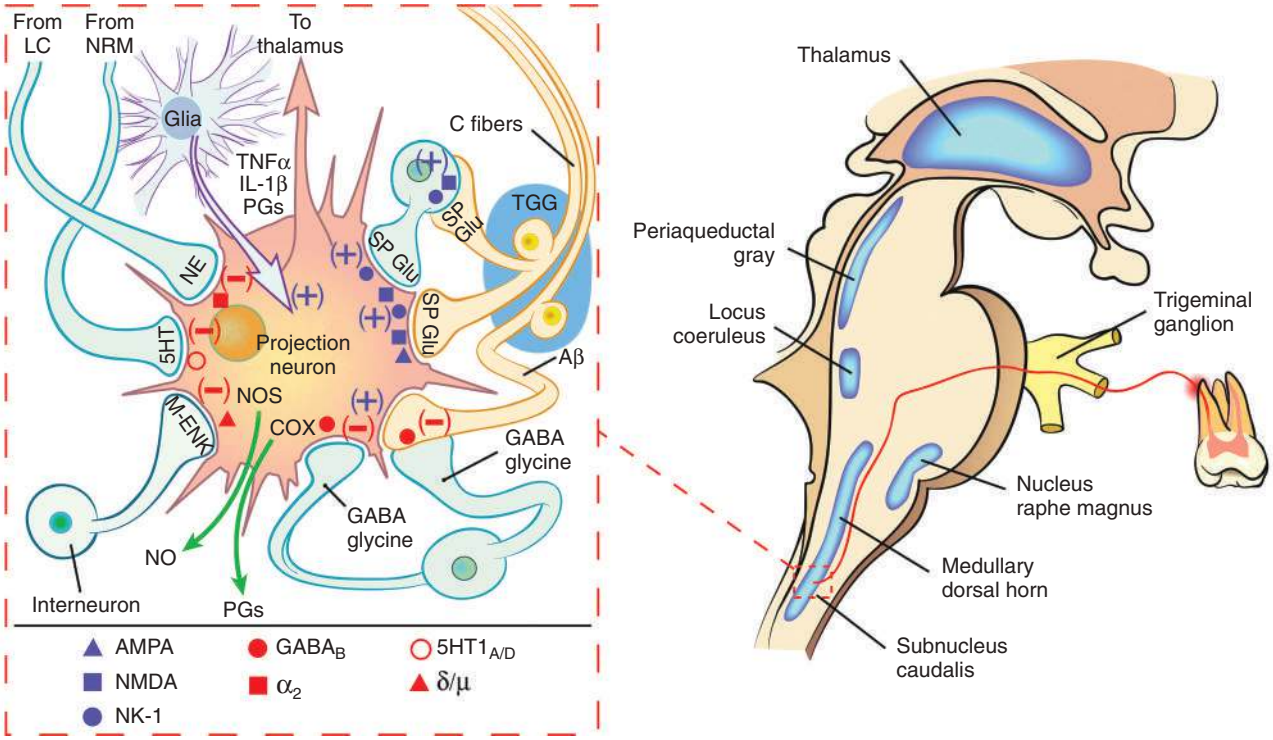


Fig. 13.32 Schematic diagram of the perception and modulation of orofacial pain. Activation of primary afferent fibers (in this example from an inflamed maxillary molar) leads to the entry of a nociceptive signal that is conveyed across a synapse in the subnucleus caudalis of the trigeminal spinal nucleus. The second-order neuron projects to the thalamus; the information is then relayed to the cortex. A great deal of processing of nociceptive input can occur at the level of the medullary dorsal horn (MDH). The *inset* depicts a typical wide dynamic range (WDR) projection neuron and its relationship with other components of the MDH. Primary afferent fibers release the excitatory amino acid, glutamate—which binds and activates either AMPA or NMDA receptors—and substance P—which activates NK-1 receptors on the WDR neuron or excitatory interneurons. Descending fibers from the locus coeruleus *(LC)* and nucleus raphe magnus *(NRM)* secrete serotonin *(5HT)* and norepinephrine *(NE)*, respectively, which inhibit transmission. Release of γ amino butyric acid *(GABA)*, the amino acid, glycine, and endogenous opioid peptides such as met-enkephalin *(M-ENK)* also inhibit transmission of nociceptive information. Projection neurons may have autocrine or paracrine effects by the synthesis and release of prostaglandins *(PGs)* and nitric oxide *(NO)* via the action of cyclooxygenase *(COX)* and nitric oxide synthase *(NOS)*, respectively. Glial cells can modulate nociceptive processing by the release of cytokines such as tumor necrosis factor α *(TNF-α)* and interleukin 1 β *(IL-1β)*. The + sign indicates an excitatory action, whereas the— sign denotes an inhibitory action. *AMPA,* α-Amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid; *NK,* neurokinin; *NMDA, N*-methyl-D-aspartate; *SP,* substance P; *TGG,* trigeminal ganglion.

caused by convergence of afferent input from different areas onto the same projection neurons.

Approximately 50% of subnucleus caudalis neurons are estimated to receive convergence of sensory input from cutaneous and deep structures.[335] In one study of a cat, a single nucleus caudalis neuron received input from sensory neurons innervating the cornea, the skin overlying the maxilla, a maxillary premolar tooth, and a mandibular canine and premolar tooth on one side.[339] Subnuclei oralis and interpolaris also receive converging input from orofacial and muscle afferents.[338] This would explain the clinical observation of patients who perceive pain in a particular tooth that actually originates from either a different tooth or structure (see also Chapter 4). In such cases, anesthetizing the tooth suspected by the patient would afford no relief. However, if an anesthetic is delivered selectively to the suspected primary source of pain, the patient's discomfort should greatly diminish.[285] Likewise, if the source of a perceived toothache were located in a muscle of mastication, palpation of that muscle should aggravate the pain.[417]

In the medullary dorsal horn, local circuit interneurons have the potential to affect the transmission of nociceptive input from primary afferents to projection neurons. Depending on the transmitter released, these neurons have the ability to enhance or diminish the signal. Typically, excitatory interneurons release glutamate or SP, whereas inhibitory interneurons release the amino acid, glycine, or γ amino butyric acid (GABA).[218,337]

The terminals of neurons that descend from brain structures such as the locus coeruleus and nucleus raphe magnus tend to inhibit nociceptive transmission at the level of the medullary dorsal horn.[14] These terminals release a variety of neuroeffective agents, including the endogenous opioid peptides (EOPs). The EOPs, similar in three-dimensional structure to many of the exogenous opiates from which their name derives, are released in response to nociceptive input and act to suppress the pain system. The EOPs likely are partly responsible for the placebo effect seen in pain control studies, because this effect can be reversed by administration of the opioid antagonist, naloxone.[141,216]

The final component of the medullary dorsal horn complex to be considered is the glial cell population. Historically considered to be solely supportive in function, they are now recognized to play an important role in the pain processing system.[308,404] Following nociceptive input from primary afferents, glia release cytokines such as TNF-α and IL-1, as well as certain PGs that may facilitate the activity of projection neurons. Glial modulating agents have been shown to be effective in experimental models of neuropathic pain,[355] and NSAIDs potentially could exert part of their analgesic mechanism by acting at this level.

Central Sensitization

Central sensitization can be defined as an increased responsiveness of central nociceptive neurons to peripheral stimulation that occurs in addition to *peripheral sensitization* of the primary afferent nociceptors. Central sensitization is thought to be a major cause of hyperalgesia and allodynia.[208] Clinical trials implicate central sensitization in patients reporting pain due to irreversible pulpitis. In one survey of nearly 1000 patients, 57% of patients with irreversible pulpitis reported mechanical allodynia (pain due to percussion).[294] This appears to be due at least in part to central sensitization, as both the ipsilateral (pulpitis) tooth and a contralateral (normal) tooth demonstrated mechanical allodynia to a force transducer.[187] Thus central sensitization contributes to a spread of endodontic pain, and the clinical application of bite force transducers may provide a novel method for diagnosing pain mechanisms.[186,187]

Different studies shed light on the molecular mechanisms involved in central sensitization (for review, see Cousins and Power[64] and Hargreaves[139]), but the process is generally initiated by a barrage of nociceptive impulses from peripheral C fibers. The level and duration of pain prior to endodontic intervention have been cited in several studies as predictors of postoperative endodontic pain,[369,400] and this may be due to such a prolonged and intense input from C nociceptors. Any reduction of such a barrage should limit the occurrence of central sensitization and the development of pain of longer duration after tissue injury (including surgical and nonsurgical endodontic procedures). The use of long-acting local anesthetics following tonsillectomies and third molar extractions has been shown to provide pain relief far beyond the duration of the peripheral tissue anesthesia.[127,169]

A reduction in the chemical mediators of inflammation at the level of the medullary dorsal horn also should reduce sensitization of second-order neurons. Decreasing the synthesis of proinflammatory PGs, cytokines, nitric oxide (NO), or the use of drugs that block the receptors of such agents probably will become accepted pharmacotherapy in the future. For example, application of an inflammatory agent to the tooth pulp of rat maxillary molars results in an increased receptive field of Aβ touch receptors on the face. This can be blocked by pretreatment with a glutamate NMDA receptor antagonist, indicating that such centrally acting drugs may offer highly efficacious means of treating odontogenic pain.[60] A similar investigation implicated NO synthesis at the level of the subnucleus caudalis in the development of tactile hypersensitivity following dental injury.[423] Reduction in NO synthase levels may also provide protection from central sensitization.[239,253]

PERCEPTION: THALAMUS TO CORTEX

The final anatomic step in the trigeminal pain pathway relies on neurons that leave the thalamus and extend to the cerebral cortex (see Fig. 13.23). The patient actually perceives a stimulus as painful at the cortical level. It is interesting to note (but likely of no surprise to the experienced clinician) that a disproportionately large portion of the sensory cortex in humans is devoted to input from orofacial regions.[300]

It is becoming increasingly obvious that higher-order (i.e., cortical) perceptual processes have a profound effect on the ultimate state of pain the patient experiences (for a review, see Yaksh[418]). Memories of previous pain experiences provide a framework by which similar new experiences are judged and serve to shape the patient's response to a given stimulus. In the field of dentistry, the anxiety level of the patient at the time of treatment has been shown to affect not only the patient's response to pain experienced during treatment,[79,414] but also the tendency of the patient to recall the experience as painful or unpleasant even

18 months after treatment.[114] The clinician should do everything possible to control a patient's anxiety level prior to endodontic treatment. One simple pretreatment method is to provide patients with positive written information regarding the control of pain during their endodontic treatment. In a placebo-controlled clinical trial, 437 endodontic patients were given one of five informative paragraphs to read prior to treatment. One of the paragraphs contained positive information about pain during treatment. Patients completed questionnaires following treatment that evaluated their dental anxiety and dental fear. Subjects given positive information were shown to be less fearful of pain during endodontic therapy.[386] Along with a positive and caring attitude, pharmacologic intervention may help reduce anxiety. Nitrous oxide has been shown to be effective in a dental setting,[76] but it may interfere with radiography procedures during endodontic therapy. In a placebo-controlled clinical trial in patients undergoing the extraction of impacted third molars, 0.25 mg of oral triazolam (a benzodiazepine) provided comparable anxiolysis to intravenous diazepam titrated to a typical clinical endpoint.[178] Of course, the patient so medicated must be provided transportation to and from the dental office, and the potential drug-drug interactions with other centrally acting agents such as opioids, barbiturates, and alcohol must be considered. One interaction that should be considered is the capacity of grapefruit juice to prolong the half-life of triazolam.[221] It has been shown that furanocoumarins in grapefruit juice inhibit cytochrome P450 3A4,[296] which is the enzyme responsible for the metabolism of triazolam in the liver. Patients should be told not to take oral triazolam with grapefruit juice.

Vascular Supply

Blood from the dental artery enters the tooth by way of arterioles having diameters of 100 μm or less. These vessels pass through the apical foramen or foramina with nerve bundles. Smaller vessels may enter the pulp by way of lateral or accessory canals. They are richly innervated by autonomic and sensory nerves, and the regulation of blood flow seems to be dominated by neuronal control (Fig. 13.33).[2,20,191,289,365]

The arterioles course up through the central portion of the radicular pulp and give off branches that spread laterally toward the odontoblast layer, beneath which they ramify to form a capillary plexus (Fig. 13.34).[200] As the arterioles pass into the coronal pulp, they fan out toward the dentin, diminish in size, and give rise to a capillary network in the subodontoblastic region (Fig. 13.35).[358] This network provides the odontoblasts with a rich source of metabolites.

Capillary blood flow in the coronal portion of the pulp is nearly twice that in the root portion.[195] Moreover, blood flow in the region of the pulp horns is greater than in all other areas of the pulp.[254] In young teeth, capillaries commonly extend into the odontoblast layer, thus ensuring an adequate supply of nutrients for the metabolically active odontoblasts (Fig. 13.36). In the subodontoblastic capillaries, fenestrations are observed in the vessel wall.[312] These fenestrations are thought to promote rapid transport of fluid and metabolites from the capillaries to the adjacent odontoblasts. The

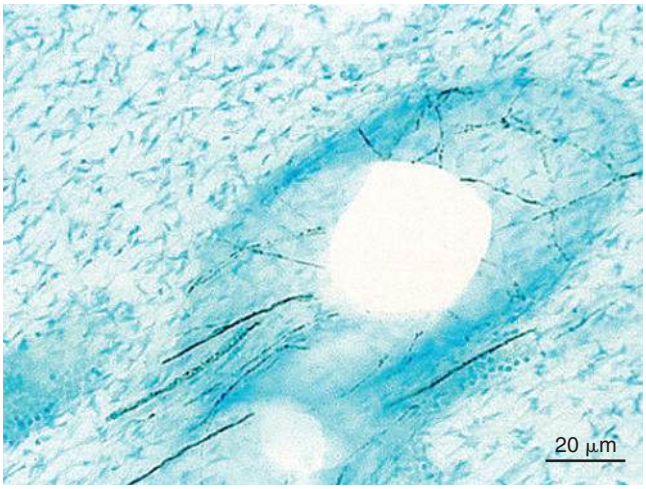


Fig. 13.33 Substance P–positive nerve fibers in the wall of pulpal blood vessels. (Courtesy Dr. K.J. Heyeraas.)

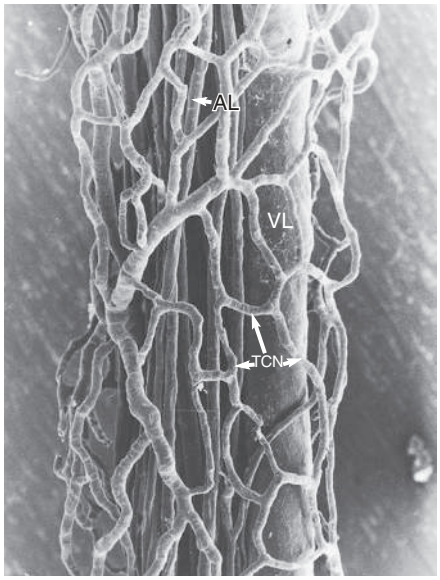


Fig. 13.34 High-power scanning electron micrograph of vascular network in the radicular pulp of a dog molar showing the configuration of the subodontoblastic terminal capillary network *(TCN)*. Venules *(VL)* and arterioles *(AL)* are indicated. (Courtesy Dr. Y. Kishi, Kanagawa Dental College, Kanagawa, Japan.)

average capillary density is about 1400/mm^2, which is greater than in most other tissues of the body.[393]

Blood passes from the capillary plexus, first into postcapillary venules (see Figs. 13.35 and 13.37) and then into larger venules.[200] Venules in the pulp have unusually thin walls, and the muscular layer is discontinuous,[70] which may facilitate the movement of fluid in or out of the vessel. The collecting venules become progressively larger as they course to the central region of the pulp. The largest venules have a diameter that may reach a maximum of 200 μm, which is considerably larger than the arterioles of the pulp.

The resting PBF is relatively high averaging 0.15 to 0.60 mL/min/g tissue.[250,367] Blood volume represents about 3% of pulpal wet weight,[31] approximately the same as in mammary tumor tissue.[410] As would be anticipated, PBF is

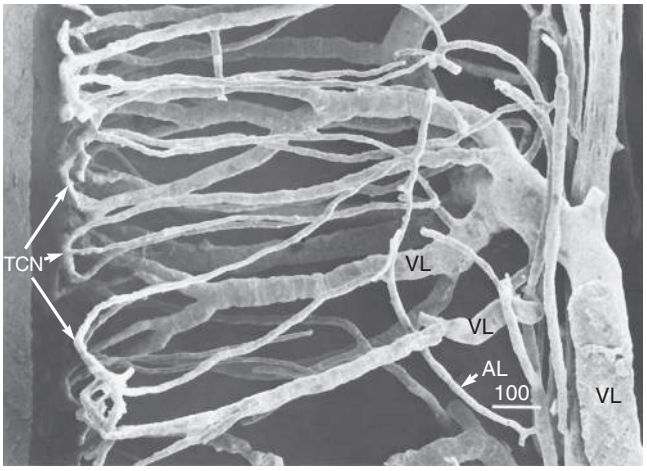


Fig. 13.35 Subodontoblastic terminal capillary network *(TCN)*, arterioles *(AL)*, and venules *(VL)* of young canine pulp. Dentin would be to the far left and the central pulp to the right. Scale bar: 100 μm. (From Takahashi K, Kishi Y, Kim S: A scanning electron microscopic study of the blood vessels of dog pulp using corrosion resin casts, *J Endod* 8:131, 1982.)

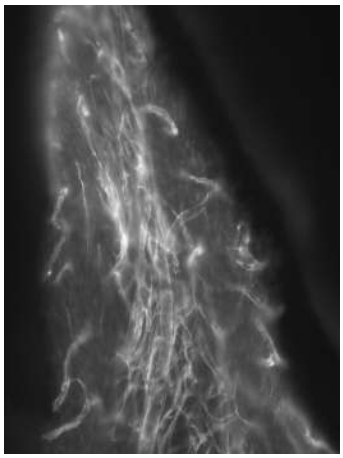


Fig. 13.36 Blood vessels in the pulp horn fan out into the odontoblast layer. (Courtesy Dr. S.R. Haug.)

greater in the peripheral layer of the pulp (i.e., the subodontoblastic capillary plexus)[200] where the oxygen consumption has been shown to be higher than in the central pulp.[26]

Changes in PBF can be measured through dentin using laser Doppler flowmeters. Sensitivity to movement requires that they are stabilized in an occlusal stent or a modified rubber dam clamp.[95,327] Because up to 80% of the Doppler signal originates from periodontal tissue, it is helpful to cover periodontal tissues with a black rubber dam.[146] Laser Doppler flowmetry can be used to detect revascularization of traumatized teeth.[80,90] Although measurement of PBF would be an ideal tool for determining pulp vitality, the use of laser Doppler and other techniques is limited due to sensitivity, specificity, reproducibility, and costs.

REGULATION OF PULPAL BLOOD FLOW

Under normal physiologic conditions, pulpal vascular tone is controlled by neuronal, paracrine, and endocrine mechanisms that keep the blood vessels in a state of partial constriction. The PBF is also influenced by vascular tone in neighboring tissues. Vasodilatation in these tissues has been shown to cause a drop in PBF due to reduction in local arterial pressure of the teeth and thereby reduced pulpal perfusion pressure.[366] The "stealing" of dental perfusion pressure makes the dental pulp vulnerable in clinical situations with inflammatory processes in the adjacent tissues, as in gingivitis and periodontitis. There is no autoregulation of PBF as it follows passively changes in systemic blood pressure.[189]

Neuronal regulation of blood flow is extensive in the pulp. There is little or no vasoconstrictor tone of sympathetic origin in the dental pulp during resting conditions,[167,365] but a neuronal vasodilator tone caused by release of sensory neuropeptides has been demonstrated (Fig. 13.38).[21,20]

There are α-adrenergic receptors in the pulp,[164] and stimulation of the cervical sympathetic trunk causes vasoconstriction and fall in PBF that can be partially reversed by α-receptor blockade.[191,365] NPY, colocalized with NE in pulpal sympathetic nerve fibers, also contributes to vasoconstriction in the pulp.[82,197]

Increase in PBF is observed after electrical tooth stimulation and is caused by the release of sensory neuropeptides followed

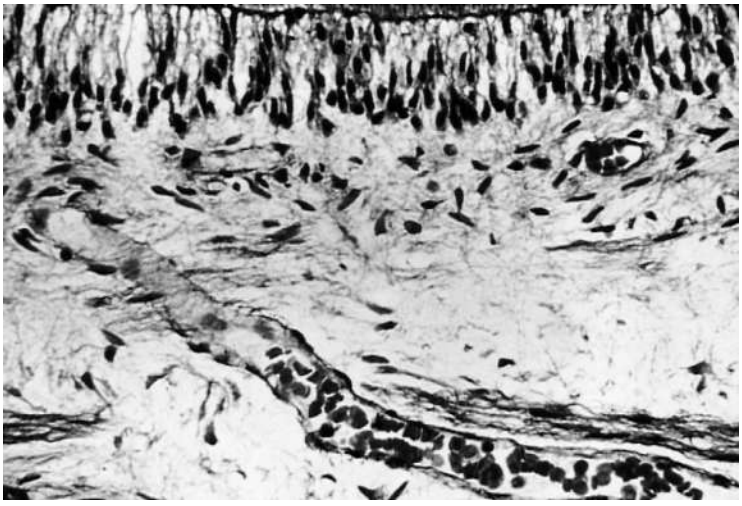


Fig. 13.37 Postcapillary venule draining blood from subodontoblastic capillary plexus.

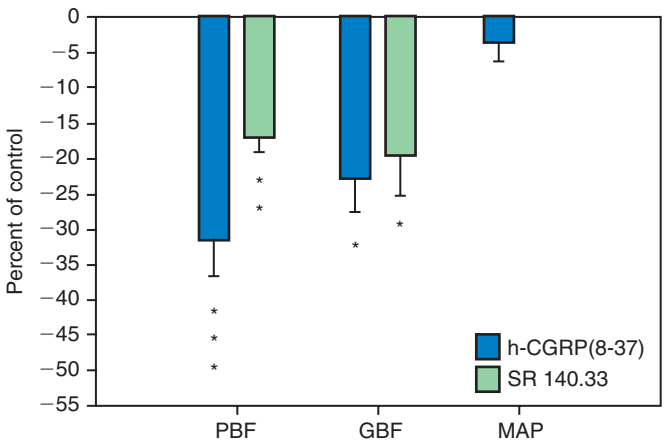


Fig. 13.38 Effect of antagonist infusion of h-*CGRP(8-37)* (calcitonin gene–related peptide inhibitor) and *SR* 140.33 (substance P inhibitor) on basal pulpal blood flow *(PBF)* and gingival blood flow *(GBF)*. *MAP,* Mean arterial pressure. ***, p <0.001, **, p <0.005, *, p <0.05, as compared with control recordings before antagonist infusion. (From Berggreen E, Heyeraas KJ: Effect of the sensory neuropeptide antagonists h-CGRP[8-37] and SR 140.33 on pulpal and gingival blood flow in ferrets, *Arch Oral Biol* 45:537, 2000.)

by vasodilatation.[20,155,183] CGRP released from sensory nerve fibers is mainly responsible for the observed vasodilatation.[20,21]

Glutamate, present in CGRP negative sensory afferent nerve fibers in the pulp, also has a vasodilatory effect when applied in the pulp during experimental conditions.[426]

There is evidence for sympathetic modulation of sensory neuropeptide release in the dental pulp[140]; presynaptic adrenoceptors are found on the sensory nerve terminals and attenuate the release of vasodilators from the sensory nerves.[35,183]

Muscarinic receptors have been identified in the pulp,[33] and the parasympathetic neurotransmitter ACh causes vasodilatation and increases blood flow in the tissue.[424] The vasodilation evoked by ACh has been demonstrated to be partly dependent on NO production.

VIP, which coexists with ACh in postganglionic neurons, is found in the dental pulp[382,397] and has been demonstrated to cause vasodilatation and increase in PBF in cats.[289]

On the other hand, Sasano and coworkers[329] failed to demonstrate parasympathetic nerve-evoked vasodilatation in the cat dental pulp, leaving pulpal vascular responses to parasympathetic neurotransmitters with some uncertainty.

Local Control of Blood Flow

The microvascular bed in the dental pulp has the ability to regulate hemodynamics in response to local tissue demands. Endothelin-1 is located in the endothelium of pulpal vasculature,[58] and close intraarterial infusions of endothelin-1 reduce PBF.[22,118,424] However, endothelin-1 does not seem to influence blood vessel vascular tone under basal, resting conditions.[22]

The endothelium in pulpal blood vessels modulates vascular tone by release of vasodilators such as prostacyclin and NO. A basal synthesis of NO provides a vasodilator tone on pulpal vessels.[20,226] The shear forces that blood flow exerts on endothelial cells seem to regulate the release of NO.[77]

Adenosine is released from ischemic and hypoxic tissue and is probably important in the metabolic regulation of blood flow in periods of low pulpal oxygen tension. When applied from the extraluminal side of the vessel wall, adenosine mediates vasodilatation in pulpal vessels.[424]

Humoral Control of Blood Flow

Evidence for humoral control of PBF exists and takes place when vasoactive substances transported by the bloodstream reach the receptors in the pulp tissue. Angiotensin II is produced by activation of the renin/angiotensin system and exerts a vasoconstrictive basal tone on pulpal blood vessels.[22] The angiotensin II receptors, AT1 and AT2, have been identified in rat pulp.[348]

Similarly to the effect of NE released from sympathetic nerve fibers in the pulp, epinephrine released from the adrenal medulla will cause vasoconstriction due to activation of α-adrenergic receptors in the pulp. Another catecholamine, dihydroxyphenylalanine (DOPA), also induces vasoconstriction in pulpal arterioles when applied intraarterially.[424]

FLUID DRAINAGE

Interstitial fluid, which accumulates in the tissue during normal conditions through net filtration out of the blood vessels or during inflammation where the net filtration is increased, must be removed in order to maintain normal fluid balance. In most tissues in the body, the lymphatic vessels drain excess fluid from the peripheral tissue and return it into the blood vessel system. In addition, the lymphatic system is important because it transports captured antigen and presents it in the lymph nodes. The existence of lymphatics in the pulp has been a matter of debate because it is difficult to distinguish between blood and lymphatic vessels by ordinary microscopic techniques without specific lymphatic markers.

Several specific lymphatic markers have now been applied, and contradictory conclusions have been drawn. One of the markers tested is vascular endothelial growth factor receptor (VEGFR-3), known to be expressed by lymphatic endothelial cells in adult tissue.[114,242] The receptor expression was reported in human and mouse pulp tissue, but to identify lymphatic vessels use of more than one lymphatic marker is recommended. VEGFR-3 has recently been demonstrated in blood vessels and on various immune cells and is not specific to lymphatic vessels.[30,390] However, other studies have failed to demonstrate lymphatic markers in the pulp [116,247] and one study showed lymphatic vessel endothelium receptor-1 (LYVE-1) staining in immune cells in human pulp, but not in vascular structures (Fig. 13.39). The same observation was done in a characterization of the lymphatic system of the mouse head.[228] The authors showed that lymphatic vessels were found in the mandibular canal. We have demonstrated lymphatic vessels in the mandibular canal in rats starting below the molars (Fig. 13.40). Taken together, we can conclude that the pulp is not supplied with draining lymphatic vessels.

Transcapillary Fluid Exchange

In all tissues in the body, the fluid transport between the blood vessels and the interstitial space is regulated by differences in colloid osmotic and hydrostatic pressures in the plasma and the interstitium, and by properties in the capillary membrane (Fig. 13.41). From the interstitium excess fluid is transported back to the blood circulation through the

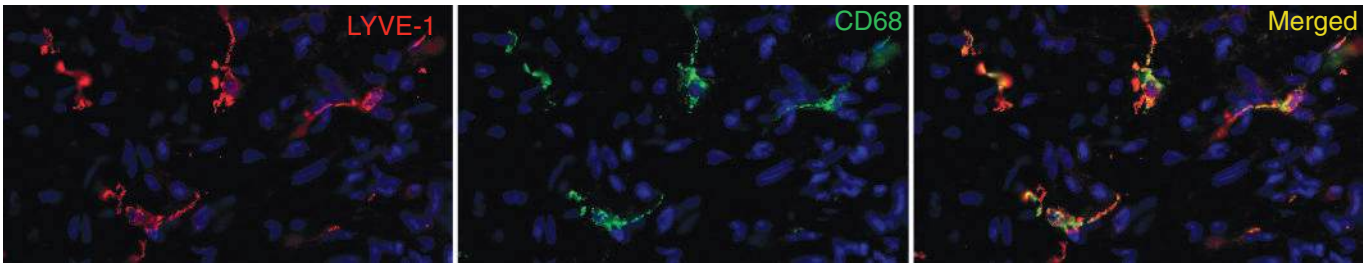


Fig. 13.39 Immune cells in normal human pulp are immunopositive to *LYVE-1*, known as a lymphatic vessel marker. The *CD68*$^+$/*LYVE-1*$^+$ cells derive from the monocytic lineage of cells. Immunostaining demonstrated the lack of *LYVE-1*$^+$ lymphatic vessels in the pulp. (Courtesy Dr. A Virtej.)

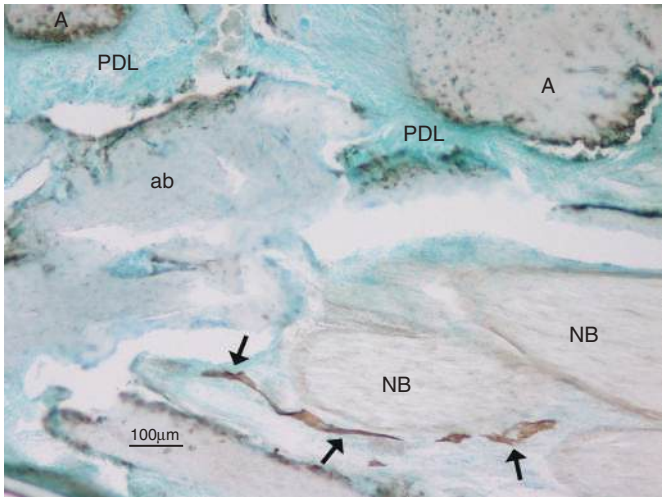


Fig. 13.40 A lymphatic vessel *(arrows)* in the mandibular canal in a rat jaw below the apex *(A)* of the molars. *ab,* Alveolar bone; *NB,* nerve bundle; *PDL,* periodontal ligament. (Courtesy Dr. Anca Virtej.)

lymphatic system. The pulp seems to be an exception along with, for example, the brain and bone marrow, as lymphatic vessels are not detected in the tissue (see the previous section). During normal conditions, a steady state is achieved as the fluid filtered into the interstitial space equals the amount of fluid transported out of the same compartment.

Using radioisotopes, the interstitial fluid volume in the pulp was measured and averaged 0.6 ± 0.03 mL/g wet weight,[31] demonstrating that as much as 60% of the extracellular fluid in dental pulp is located outside the vascular system. Measurements of interstitial fluid pressure in the pulp with the micropuncture method have given values that range from 6 to 10 mm Hg,[20,152] but higher values measured with different methods have also been reported.[40,385,392]

Colloid osmotic pressure (COP) measurements in interstitial fluid isolated from rat incisors have shown a relatively high pulpal COP, reaching 83% of plasma COP.[31] The high value may imply that the normal permeability of pulpal vessels to plasma proteins is relatively high or the drainage of plasma proteins is ineffective.

Because lymphatic vessels are lacking inside the pulp, excess interstitial fluid and proteins must be transported out of the pulp by other transport routes in order to achieve a steady-state situation. Two possibilities exists: (1) transport of fluid in the interstitial compartment toward the apical part of the pulp and furthermore out of the apex, and (2) a combination of fluid reabsorption into pulpal blood vessels in addition to transport of protein-rich fluid toward the apex.

CIRCULATION IN THE INFLAMED PULP

Inflammation in the pulp takes place in a low-compliance environment composed of rigid dentinal walls. *Compliance*

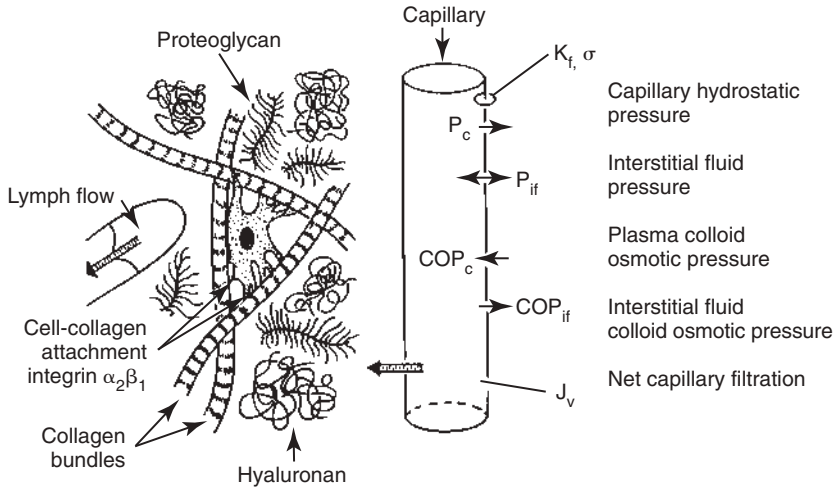


Fig. 13.41 Interstitial structure and pressures that govern transcapillary fluid transport. K_f, Capillary filtration coefficient; σ, capillary reflection coefficient for plasma proteins. (From Wiig H, Rubin K, Reed RK: New and active role of the interstitium in control of interstitial fluid pressure: potential therapeutic consequences, *Acta Anaesthesiol Scand* 47:111, 2003.)

is defined as the relationship between volume (V) and interstitial pressure (P) changes: $C = \Delta V/\Delta P$. Consequently, in the low-compliant pulp, an increase in blood or interstitial volume will lead to a relatively large increase in the hydrostatic pressure in the pulp. The acute vascular reactions to an inflammatory stimulus are vasodilatation and increased vascular permeability, both of which will increase pulp interstitial fluid pressure[153,156,365,384] and may tend to compress blood vessels and counteract a beneficial blood flow increase (Fig. 13.42).

Classical studies have demonstrated that an increase in intrapulpal tissue pressure promoted absorption of tissue fluid back into the circulation, thereby reducing the pressure.[153,156] This observation can explain why pulpal tissue pressure in inflamed pulps may persist in local regions for long observation periods,[367] contradicting the old concept of a wide, generalized collapse of pulpal venules and cessation of blood flow (pulpal strangulation theory).

The delivery of dental restorative procedures may lead to substantial increases or decreases in PBF, depending on the precise procedure and time point sampled.[191] Vasoactive mediators are locally released upon an inflammatory insult, and in the pulp, PGE_2, BK, SP, and histamine have all been demonstrated to increase PBF after application.[194,290] In contrast, serotonin (5-HT) is released primarily from the platelets, and given intraarterially, it has been shown to reduce PBF.[196,424]

Acute inflammation in the dental pulp induces an immediate rise in blood flow and can reach a magnitude of up to nearly 200% of control flow followed by increased vascular permeability.[154,156]

A common outcome of pulpal inflammation is development of tissue necrosis. One study found circulatory dysfunction developed in the pulp after exposure to lipopolysaccharide (LPS) from gram-negative bacteria.[31]

In addition, the inflammatory cytokines IL-1 and TNF-α are elevated in the inflamed pulp. When the endothelium is exposed to endotoxin, it expresses cytokines, chemokines, and thromboxane A2. The latter has been demonstrated to be produced in the pulp exposed to LPS[287] and induces vasoconstriction. The set of changes in endothelial function have been called *endothelial perturbation* and were first described in endothelial cells exposed to endotoxin or to cytokines such as IL-1, TNF-α, and IL-6.[24,276] The activated endothelium also participates in procoagulant reactions that promote fibrin clot formation.[352] A reduced pulpal perfusion due to endothelial perturbation might be the consequence of bacterial infection impairing pulpal defense mechanisms and promoting necrosis. Downregulation of vascular endothelial growth factor (VEGF) expression in stromal cells and reduced microvessel density have been observed in human dental pulps with irreversible pulpitis.[10] VEGF is an essential proangiogenic factor, and the reduced microvessel density might also lead to reduced pulpal perfusion and contribute to development of pulpal necrosis.

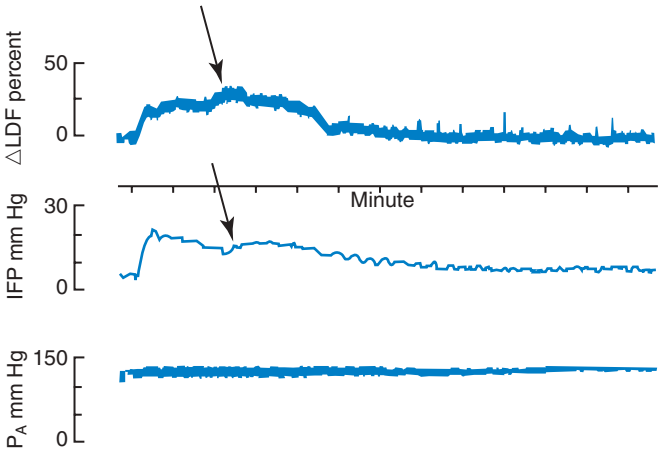


Fig. 13.42 Original simultaneous recordings of percent change in pulpal blood flow (ΔLDF%), interstitial fluid pressure *(IFP)*, and systemic blood pressure *(P_A mm Hg)* in a cat during electrical tooth stimulation. Note that when IFP is first decreasing after an initial rise, the pulpal blood flow reaches its maximal level *(arrows)*, demonstrating compression of vessels in the first phase. (Courtesy Dr. K.J. Heyeraas.)

Vascular Permeability

Increased vascular permeability takes place as a result of acute inflammation, and vascular leakage has been demonstrated in the pulp after release of inflammatory mediators such as PG, histamine, BK, and the sensory neuropeptide, SP.[184,194,241]

LPS and lipoteichoic acid from gram-negative and gram-positive bacteria, respectively, cause upregulation of VEGF in activated pulpal cells.[34,360] VEGF increases vascular permeability,[93,334] and it is likely that it also causes leakage in pulpal vessels. It is a potent agent, because its ability to enhance microvascular permeability is estimated to be 50,000 times higher than that of histamine.[341] Cytokines such as IL-1 and TNF-α are released into the pulp interstitial fluid during inflammation[31] and upregulate VEGF mRNA gene expression in pulpal fibroblasts.[61]

The resulting increased vascular permeability allows increased transport of proteins through the capillary vessel wall and results in increased COP in the tissue. In acute pulpitis induced by LPS, it has been shown that COP in the pulp can reach the level of plasma COP, meaning that a protein transport barrier between plasma and interstitium can be eliminated.[31]

The lack of lymphatic vessels in the pulp may in this situation disfavor the pulps ability to recover and reestablish a normal COP since the transport of proteins is dependent on a pressure difference for fluid transport out of the pulp. Lack of lymphatic vessel for transport of protein-rich fluid may make the pulp more vulnerable to development of necrosis after severe pulpitis.

Clinical Aspects

The influence of posture on PBF has been observed in humans.[95] Significantly greater PBF was measured when subjects changed from an upright to a supine position. The supine position increases venous return from all tissues below the level of the heart, thereby increasing cardiac output and producing a transient increase in systemic blood pressure. The increase in blood pressure stimulates baroreceptors that reflexively decrease sympathetic vasoconstriction to all vascular beds, thereby increasing peripheral blood flow. Patients with pulpitis often report an inability to sleep at night because they are disturbed by throbbing tooth pain. In addition to the lack of distractions normally present during the day, the following mechanism may be operative in patients with

inflamed pulps. When these patients lie down at the end of the day, their PBF probably increases due to the cardiovascular postural responses described earlier. This may increase their already elevated pulpal tissue pressure,[153,156,351,367,384] which is then sufficient to activate sensitized pulpal nociceptors and initiate spontaneous pulpal pain. Thus the "throbbing" sensation of toothache was previously considered as an effect of the pulsation in the pulp that follows heart contractions (systole), but a study investigating the rhythm of toothache and the patient pulse showed lack of synchrony between the two parameters.[256] The authors raised an alternative hypothesis: that the throbbing quality is not a primary sensation but rather an emergent property, or perception, whose "pacemaker" lies within the CNS.

Pulpal Repair

The inherent healing potential of the dental pulp is well recognized. As in all other connective tissues, repair of tissue injury commences with débridement by macrophages, followed by proliferation of fibroblasts, capillary buds, and the formation of collagen. Local circulation is of critical importance in wound healing and repair. An adequate supply of blood is essential to transport immune cells into the area of pulpal injury and to dilute and remove deleterious agents from the area. It is also important to provide fibroblasts with nutrients from which to synthesize collagen. Unlike most tissues, the pulp has essentially no collateral circulation, making it theoretically more vulnerable than most other tissues. In the case of severe injury, healing would be impaired in teeth with a limited blood supply. It seems reasonable to assume that the highly cellular pulp of a young tooth, with a wide-open apical foramen and rich blood supply, has a much better healing potential than an older tooth with a narrow foramen and a restricted blood supply.

Dentin can be classified as primary, secondary, or tertiary, depending on when it was formed. *Primary dentin* is the regular tubular dentin formed before eruption, including mantle dentin. *Secondary dentin* is the regular circumferential dentin formed after tooth eruption, whose tubules remain continuous with that of primary dentin. *Tertiary dentin* is the irregular dentin that is formed in response to abnormal stimuli, such as excess tooth wear, cavity preparation, restorative materials, and caries.[65,66] In the past, tertiary dentin has been called *irregular dentin, irritation dentin, reparative dentin,* and *replacement dentin*. Much of the confusion was caused by a lack of understanding of how tertiary dentin is formed.

If the original odontoblasts that made *secondary dentin* are responsible for focal tertiary dentin formation, that particular type of tertiary dentin is termed *reactionary dentin*.[345] Generally, the rate of formation of dentin is increased, but the tubules remain continuous with the secondary dentin.[349] However, if the provoking stimulus caused the destruction of the original odontoblasts, the new, less tubular, more irregular dentin formed by newly differentiated odontoblast-like cells is called *reparative dentin*. In this dentin, the tubules are usually not continuous with those of secondary dentin. Initially, the newly formed cells tend to be cuboidal in shape, without the odontoblast process that is necessary to form dentinal tubules. They seem to form in response to the release of growth factors that were bound to collagen during the formation of secondary dentin.[94,315,345] The loss of the continuous layer of odontoblasts exposes unmineralized predentin that is thought to contain both soluble and insoluble forms of TGF-β, insulin-like growth factor (IGF)-1 and IGF-2, bone morphogenetic proteins (BMPs), VEGF, and other growth factors that attract and cause proliferation and differentiation of MSC to form reparative dentin and new blood vessels. During caries progression, bacterial acids may solubilize these growth factors from mineralized dentin, liberating them to diffuse to the pulp, where they could stimulate reactionary dentin formation. This is also thought to be the mechanism of action of calcium hydroxide during apexification treatment. Despite its high pH, calcium hydroxide has a slight demineralizing effect on dentin and has been shown to cause the release of TGF-β.[344] TGF-β and other growth factors stimulate and accelerate reparative dentinogenesis. Attempts have been made to apply growth factors to dentin to allow it to diffuse through the tubules to the pulp.[322,323,346] Although this has been successful, the remaining dentin thickness must be so thin that this approach may not be practical from a therapeutic perspective. Others have inserted deoxyribonucleic acid (DNA)-sequenced BMP-7 into retroviruses to transfect ferret pulpal fibroblasts to stimulate increased BMP-7 production. Although this was successful in normal pulps,[322] it was unsuccessful in inflamed pulps.[321] Specific amelogenin gene splice products, A + 4 and A − 4, adsorbed onto agarose beads and applied to pulp exposures, induced complete closure and mineralization of the root canal in rat molars.[124] The regulation of peritubular dentin formation is not fully understood. Some have claimed that this is a passive process resulting in occlusion of the tubules over time, but it has also been claimed that this is a mechanism under odontoblast control. If odontoblasts could be stimulated to form excessive peritubular dentin by the application of an appropriate biologic signaling molecule to the floor of cavity preparations, the tubules of the remaining dentin could be occluded, rendering this dentin impermeable and protecting the pulp from the inward diffusion of noxious substances that might leak around restorations.[297] These are examples of how molecular biology may be used in future restorative dentistry.

The term most commonly applied to irregularly formed dentin is *reparative dentin,* presumably because it frequently forms in response to injury and appears to be a component of the reparative process. It must be recognized, however, that this type of dentin has also been observed in the pulps of normal, unerupted teeth without any obvious injury.[281]

It should be recalled that secondary dentin is deposited circumpulpally at a slow rate throughout the life of the vital tooth.[349] In contrast, when a carious lesion has invaded dentin, the pulp usually responds by depositing a layer of tertiary dentin over the dentinal tubules of the primary or secondary dentin that communicate with the carious lesion (Fig. 13.43). Similarly, when occlusal wear removes the overlying enamel and exposes the dentin to the oral environment, tertiary dentin is deposited on the pulpal surface of the exposed dentin. Thus the formation of tertiary dentin allows the pulp to retreat behind a barrier of mineralized tissue.[375]

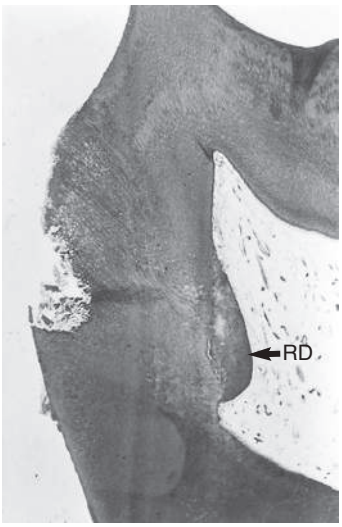


Fig. 13.43 Reparative dentin *(RD)* deposited in response to a carious lesion in the dentin. (From Trowbridge HO: Pathogenesis of pulpitis resulting from dental caries, *J Endod* 7:52, 1981.)

Compared with primary or secondary dentin, tertiary dentin tends to be less tubular, and the tubules tend to be more irregular with larger lumina. In some cases, particularly when the original odontoblasts are destroyed, no tubules are formed. The cells that form reparative dentin are often cuboidal and not as columnar as the primary odontoblasts of the coronal pulp (Fig. 13.44). The quality of tertiary dentin (i.e., the extent to which it resembles primary or secondary dentin) is quite variable. If irritation to the pulp is relatively mild, as in the case of a superficial carious lesion, then the tertiary dentin formed may resemble primary dentin in terms of tubularity and degree of mineralization. On the other hand, dentin deposited in response to a deep carious lesion may be relatively atubular and poorly mineralized, with many areas of interglobular dentin. The degree of irregularity of this dentin is probably determined by numerous factors, such as the amount of inflammation present, the extent of cellular injury, and the state of differentiation of the replacement odontoblasts.

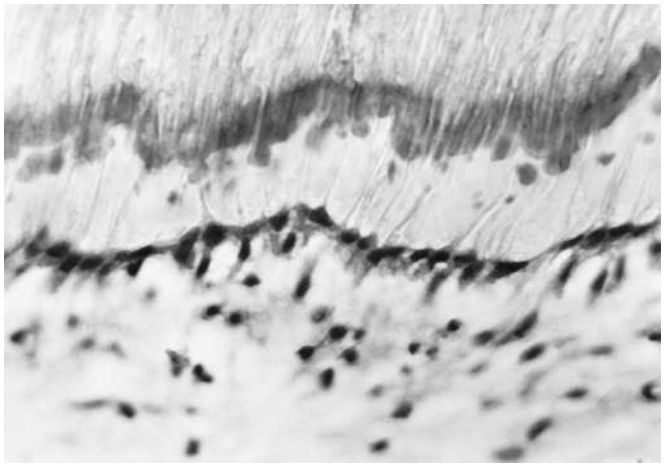


Fig. 13.44 Layer of cells forming reparative dentin. Note the decreased tubularity of reparative dentin compared with the developmental dentin above it.

The poorest quality of reparative dentin is usually observed in association with marked pulpal inflammation.[65,375] In fact, the dentin may be so poorly organized that areas of soft tissue are entrapped within the dentinal matrix. In histologic sections, these areas of soft-tissue entrapment impart a Swiss-cheese appearance to the dentin (Fig. 13.45). As the entrapped soft tissue degenerates, products of tissue degeneration are released that further contribute to the inflammatory stimuli assailing the pulp.[375]

It has been reported that trauma caused by cavity preparation that is too mild to result in the loss of primary odontoblasts does not lead to reparative dentin formation, even if the cavity preparation is relatively deep.[74] This has been confirmed in both rat[52] and human teeth.[261] However, chronic pulpal inflammation associated with deep caries produces reparative dentin. This reparative dentin is formed by new odontoblast-like cells. For many years, it has been recognized that destruction of primary odontoblasts is soon followed by increased mitotic activity within fibroblasts of the subjacent cell-rich zone. It has been shown that the progeny of these dividing cells differentiate into functioning odontoblasts.[100] Investigators[421] have studied dentin bridge formation in healthy teeth of dogs and found that pulpal fibroblasts appeared to undergo dedifferentiation and revert to undifferentiated MSC (Fig. 13.46). The similarity of primary odontoblasts to replacement odontoblasts was established by D'Souza and colleagues.[78] They were able to show that cells forming reparative dentin synthesize type I (but not type III) collagen, and they are immunopositive for dentin sialoprotein.

Destruction of primary odontoblasts can occur from cutting cavity preparations dry,[81,206] from bacterial products such as endotoxins shed from deep carious lesions,[18,402] or from mechanical exposure of pulps.[262] Such pulpal wounds do not heal if the tissue is irreversibly inflamed.[65] Local fibroblast-like cells divide, and the new cells then differentiate into new odontoblasts. Recalling the migratory potential of

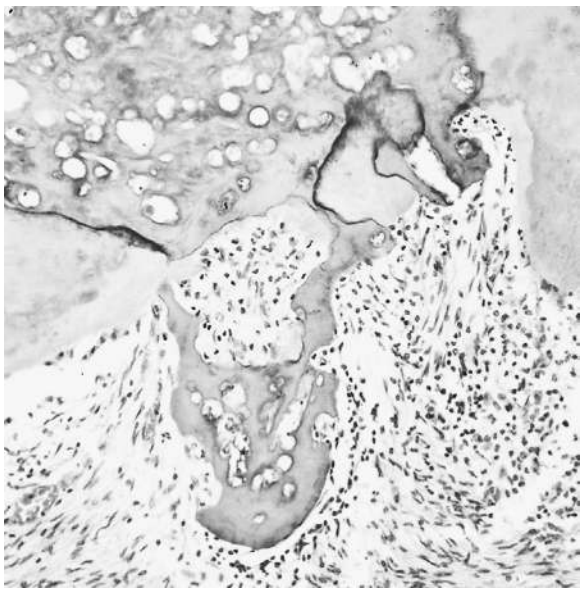


Fig. 13.45 Swiss-cheese type of reparative dentin. Note the numerous areas of soft-tissue inclusion and infiltration of inflammatory cells in the pulp.

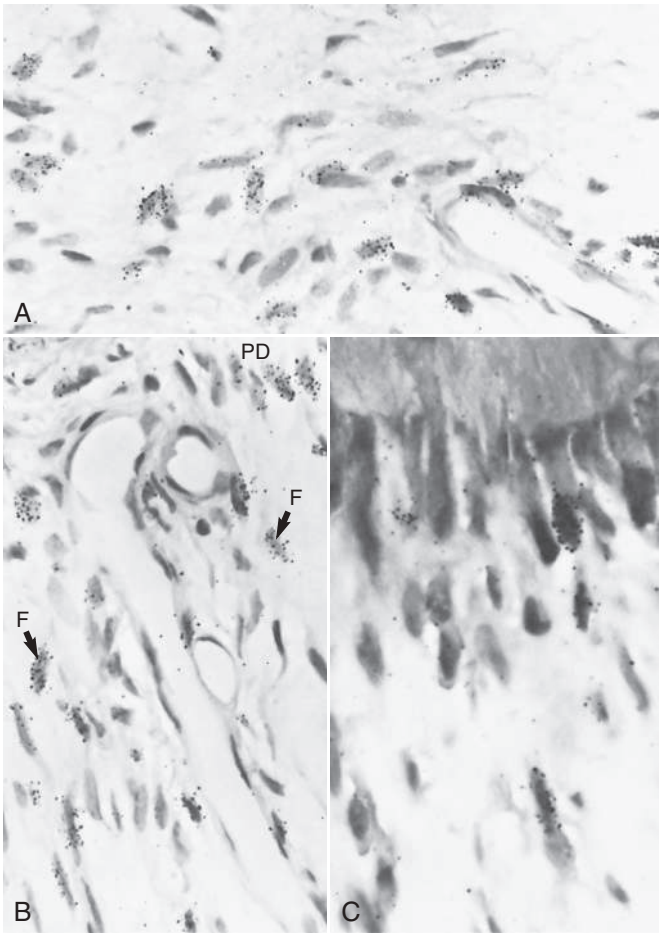


Fig. 13.46 Autoradiographs from dog molars illustrating uptake of ^{3}H-thymidine by pulp cells preparing to undergo cell division after pulpotomy and pulp capping with calcium hydroxide. **A,** Two days after pulp capping. Fibroblasts, endothelial cells, and pericytes beneath the exposure site are labeled. **B,** By the fourth day, fibroblasts *(F)* and preodontoblasts adjacent to the predentin *(PD)* are labeled, which suggests that differentiation of preodontoblasts occurred within 2 days. **C,** Six days after pulp capping, new odontoblasts are labeled, and tubular dentin is being formed. (Titrated thymidine was injected 2 days after the pulp capping procedures in **[B]** and **[C].**) (From Yamamura T, Shimono M, Koike H, et al: Differentiation and induction of undifferentiated mesenchymal cells in tooth and periodontal tissue during wound healing and regeneration, *Bull Tokyo Dent Coll* 21:181, 1980.)

ectomesenchymal cells from which the pulpal fibroblasts are derived, it is not difficult to envision the differentiating odontoblasts moving from the subodontoblastic zone to the area of injury to constitute a new odontoblast layer. Activation of antigen-presenting dendritic cells by mild inflammatory processes may also promote osteoblast/odontoblast-like differentiation and expression of molecules implicated in mineralization. Recognition of bacteria by specific odontoblast and fibroblast membrane receptors triggers an inflammatory and immune response within the pulp tissue that would also modulate the repair process.[123]

Although many animal studies have shown dentin bridge formation in healthy pulps following pulp capping with adhesive resins,[65] such procedures fail in normal human teeth.[62]

When small mechanical pulp exposures are inadvertently made in healthy teeth, the recommendation has been to place a small, calcium hydroxide–containing dressing on the wound. After setting, the surrounding dentin can be bonded using a no-rinse, self-etching primer adhesive.[177] Calcium silicate cements like mineral trioxide aggregate (MTA) have also been recognized to promote hard-tissue formation, and available information indicates that the dentin bridge formed under these cements is more dense and has fewer defects compared with traditional calcium hydroxide–containing dressings.[3,5,266,373]

The formation of atubular "fibrodentin" is another potential product of newly differentiated odontoblasts, provided that a capillary plexus develops beneath the fibrodentin.[15] This is consistent with the observation that the newly formed dentin bridge is composed first of a thin layer of atubular dentin on which a relatively thick layer of tubular dentin is deposited.[65,100] The fibrodentin was lined by cells resembling mesenchymal cells, whereas the tubular dentin was associated with cells closely resembling odontoblasts.

Researchers have also studied reparative dentin formed in response to relatively traumatic experimental class V cavity preparations in human teeth.[349] They found that seldom was reparative dentin formed until about the 30th postoperative day. The rate of dentin formation was 3.5 μm/day for the first 3 weeks after the onset of dentinogenesis, after which it decreased markedly. By postoperative day 132, dentin formation had nearly ceased. Assuming that most of the odontoblasts were destroyed during traumatic cavity preparation, as was likely in this experiment, the 30-day delay between cavity preparation and the onset of reparative dentin formation is thought to reflect the time required for the proliferation, migration, and differentiation of new replacement odontoblasts.

Does reparative dentin protect the pulp, or is it simply a form of scar tissue? To serve a protective function, it would have to provide a relatively impermeable barrier that would exclude irritants from the pulp and compensate for the loss of developmental dentin. The junction between developmental and reparative dentin has been studied using a dye diffusion technique, which demonstrated the presence of an atubular zone situated between secondary dentin and reparative dentin (Fig. 13.47).[96] In addition to a dramatic



Fig. 13.47 Diffusion of dye from the pulp into reparative dentin. Note atubular zone between reparative dentin (RD) and primary dentin on the left. (From Fish EW: *Experimental investigation of the enamel, dentin, and dental pulp,* London, 1932, John Bale Sons & Danielson.)

reduction in the number of tubules, the walls of the tubules along the junction were often thickened and occluded with material similar to peritubular matrix.[332] Taken together, these observations would indicate that the junctional zone between developmental and reparative dentin is an atubular zone of low permeability. Moreover, the accumulation of pulpal dendritic cells was reduced after reparative dentin formation, which may indicate the reduction of incoming bacterial antigens.[325]

Periodontally diseased teeth have smaller root canal diameters than teeth that are periodontally healthy.[207] The root canals of such teeth are narrowed by the deposition of large quantities of reactionary dentin along the dentinal walls.[333] The decrease in root canal diameter with increasing age, in the absence of periodontal disease, is more likely to be the result of secondary dentin formation.

One study showed that in a rat model, frequent scaling and root planing resulted in reparative dentin formation along the pulpal wall subjacent to the instrumented root surface.[148] However, given that normal rat root dentin is only 100 μm thick, these procedures are probably more traumatic to the pulp in the rat model than in humans, where normal root dentin is more than 2000 μm thick.

Not uncommonly, the cellular elements of the pulp are largely replaced by fibrous connective tissue over a span of five decades. It appears that in some cases, the pulp responds to noxious stimuli by accumulating large fiber bundles of collagen, rather than by elaborating reparative dentin (Fig. 13.48). However, fibrosis and reparative dentin formation often go hand in hand, indicating that both are expressions of a reparative potential. In periodontally diseased teeth, the pulp tissue is found to be the site of an enhanced process of collagenous fibrosis associated with an inflammatory infiltrate.[57]

With the expanding knowledge of tooth regeneration and biologic mechanisms of functional dental tissue repair, current treatment strategies are beginning to give way to evolving fields such as tissue engineering and biomimetics. Pulpal stem cells in scaffolds have been shown to produce pulplike tissues with tubular-like dentin,[89] and in animal models, root perforations have been treated with scaffolds of collagen, pulpal stem cells, and dentin matrix protein 1, resulting in organized matrix similar to that of pulpal tissue.[310]

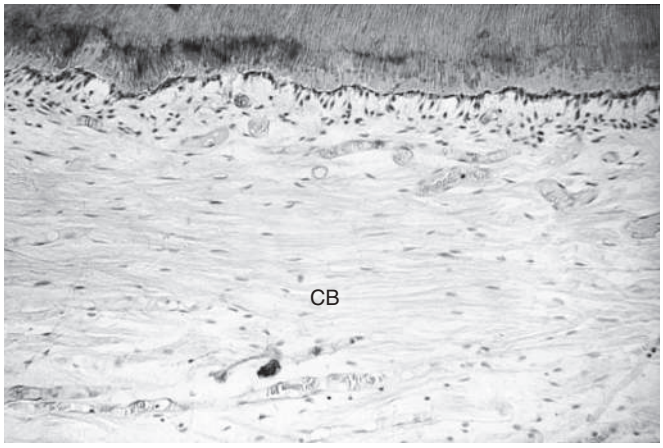


Fig. 13.48 Fibrosis of dental pulp showing replacement of pulp tissue by large collagen bundles *(CB)*.

Studies investigating new possibilities for regeneration of the pulp/dentin complex are now frequently reported and are described in detail in Chapter 12. A first interesting case report, which has been followed up by others, has led to new strategies for treatment of necrotic immature roots (Fig. 13.49).[13] In the future, the field of pulpal repair will probably develop rapidly, and new treatment strategies will appear.

Pulpal Calcifications

The calcification of pulp tissue is a common occurrence. Although estimates of the incidence of this phenomenon vary widely, it is safe to say that one or more pulp calcifications are present in at least 50% of all teeth. In the coronal pulp, calcification usually takes the form of discrete, concentric pulp stones (Fig. 13.50), whereas in the radicular

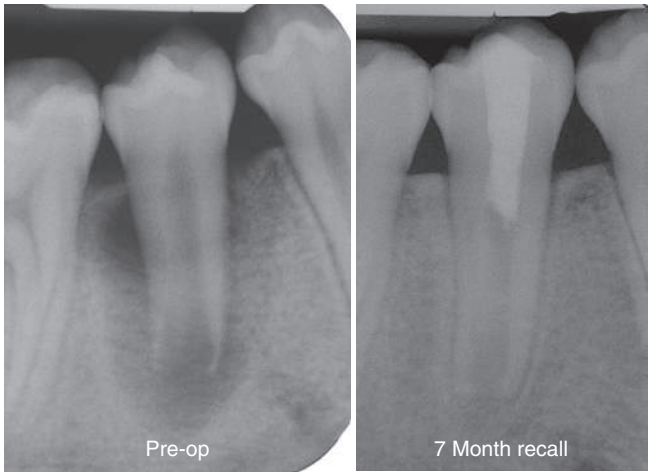


Fig. 13.49 Immature tooth with a necrotic infected canal with apical periodontitis. The canal is disinfected with copious irrigation with sodium hypochlorite and an antibiotic paste. Seven months after treatment, the patient is asymptomatic, and the apex shows healing of the apical periodontitis and some closure of the apex. (From Banchs F, Trope M: Revascularization of immature permanent teeth with apical periodontitis: new treatment protocol? *J Endod* 30:196, 2004.)

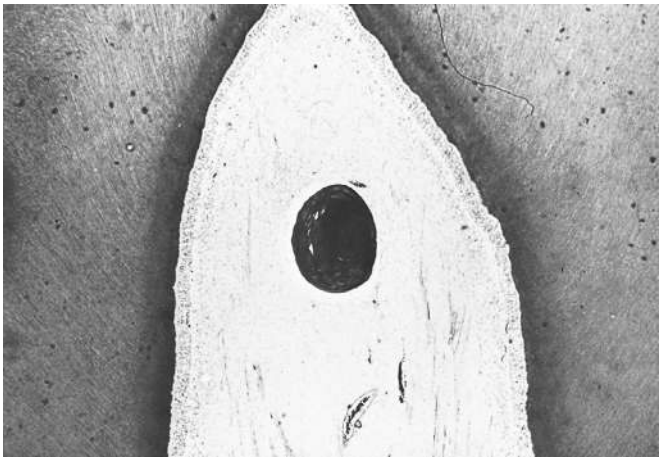


Fig. 13.50 Pulp stone with a smooth surface and concentric laminations in the pulp of a newly erupted premolar extracted in the course of orthodontic treatment.

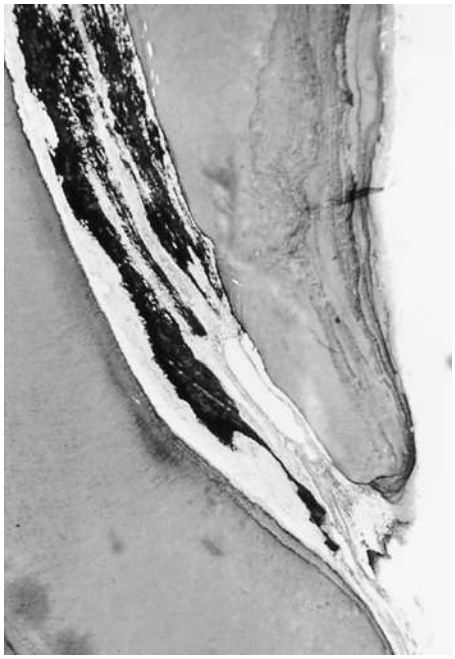


Fig. 13.51 Diffuse calcification near the apical foramen.

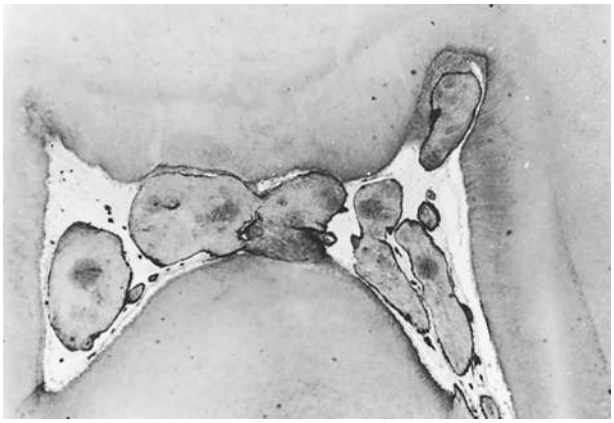


Fig. 13.52 Pulp stones occupying much of the pulp chamber.

pulp, calcification tends to be diffuse (Fig. 13.51).[379] There is no clear evidence as to whether pulp calcification is a pathologic process related to various forms of injury or a natural phenomenon. The clinical significance of pulp calcification is that it may hinder root canal treatment.

Pulp stones (denticles) range in size from small, microscopic particles often seen in association with the wall of arterioles to accretions that occupy almost the entire pulp chamber (Fig. 13.52). The mineral phase of pulp calcifications has been shown to consist of typical carbonated hydroxyapatite.[379] Histologically, two types of stones are recognized: (1) those that are round or ovoid, with smooth surfaces and concentric laminations (see Fig. 13.50), and (2) those that assume no particular shape, lack laminations, and have rough surfaces (Fig. 13.53). Laminated stones appear to grow by the addition of collagen fibrils to their surface, whereas unlaminated stones develop by way of the mineralization of preformed collagen fiber bundles. In the latter type, the mineralization front seems to extend out along the coarse fibers, making the surface of the stones appear fuzzy (Fig. 13.54). Often these coarse fiber bundles appear to have undergone hyalinization, thus resembling old scar tissue.

Pulp stones may also form around epithelial cells (i.e., remnants of Hertwig epithelial root sheath). Presumably the epithelial remnants induce adjacent MSC to differentiate into odontoblasts. Characteristically these pulp stones are found near the root apex and contain dentinal tubules (see Fig. 13.19).

The cause of pulpal calcification is largely unknown. Calcification may occur around a nidus of degenerating cells, blood thrombi, or collagen fibers. Many authors believe that this represents a form of dystrophic calcification. In this type of calcification, calcium is deposited in tissues that are degenerating. Calcium phosphate crystals may be deposited within the cells themselves. Initially this takes place within the mitochondria because of the increased membrane permeability to calcium resulting from a failure to maintain active transport systems within the cell membranes. Thus degenerating cells serving as a nidus may

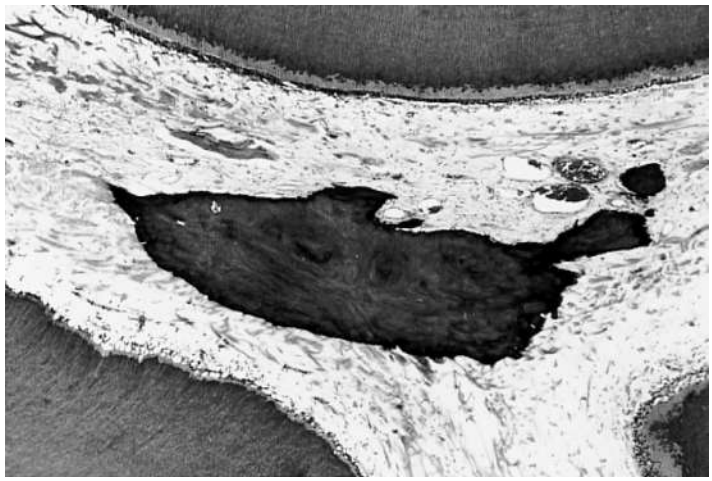


Fig. 13.53 Rough surface form of pulp stone. Note hyalinization of collagen fibers.

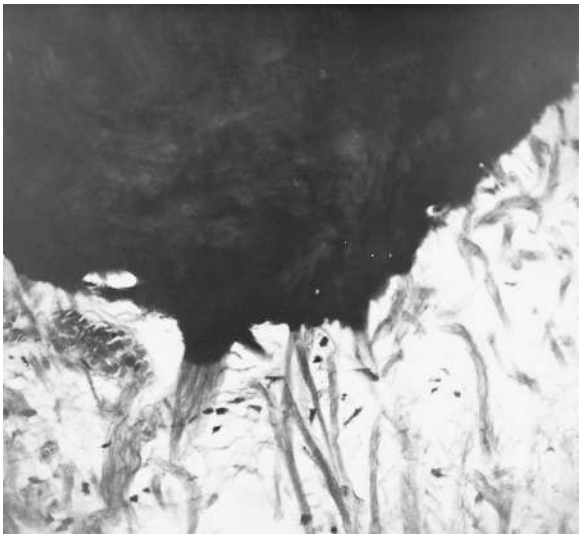


Fig. 13.54 High-power view of a pulp stone from Fig. 13.53, showing the relationship of mineralization fronts to collagen fibers.

initiate calcification of a tissue. In the absence of obvious tissue degeneration, the cause of pulpal calcification is enigmatic. It is often difficult to assign the term *dystrophic calcification* to pulp stones because they so often occur in apparently healthy pulps, suggesting that functional stress need not be present for calcification to occur. Calcification in the mature pulp is often assumed to be related to the aging process, but in a study involving 52 impacted canines from patients between 11 and 76 years of age, there was a constant incidence of concentric denticles for all age groups, indicating no relation to aging.[281] Diffuse calcifications, on the other hand, increased in incidence to age 25 years; thereafter they remained constant in successive age groups.

At times, numerous concentric pulp stones with no apparent cause are seen in all the teeth of young individuals. In such cases, the appearance of pulp stones may be ascribed to individual biologic characteristics (e.g., tori, cutaneous nevi).[281]

Although soft-tissue collagen does not usually calcify, it is common to find calcification occurring in old hyalinized scar tissue in the skin. This may be due to the increase in the extent of cross-linking between collagen molecules (because increased cross-linkage is thought to enhance the tendency for collagen fibers to calcify). A relationship may exist between pathologic alterations in collagen molecules within the pulp and pulpal calcification.

Calcification replaces the cellular components of the pulp and may possibly hinder the blood supply, although concrete evidence for this strangulation theory is lacking. Idiopathic pulpal pain was classically attributed to the presence of pulp stones. Modern knowledge of mechanisms of nociceptor activation coupled with the observation that pulp stones are so frequently observed in teeth lacking a history of pain have largely discounted this hypothesis. Therefore from a clinical perspective, it would be unlikely that a patient's unexplained pain symptoms are due to pulpal calcifications, no matter how dramatic they may appear on a radiograph.

Luxation of teeth as a result of trauma may result in calcific metamorphosis, a condition that can, in a matter of months or years, lead to partial or complete radiographic obliteration of the pulp chamber. The cause of radiographic obliteration is excessive deposition of mineralized tissue resembling cementum or, occasionally, bone on the dentin walls, also referred to as *internal ankylosis* (Fig. 13.55). Histologic examination invariably reveals the presence of some soft tissue, and cells resembling cementoblasts can be observed lining the mineralized tissue. This calcific metamorphosis of the pulp has also been reported in replanted teeth of the rat.[280]

Clinically, the crowns of teeth affected by calcific metamorphosis may show a yellowish hue compared with adjacent normal teeth. This condition usually occurs in teeth with incomplete root formation. Trauma results in the disruption of blood vessels entering the tooth, thus producing

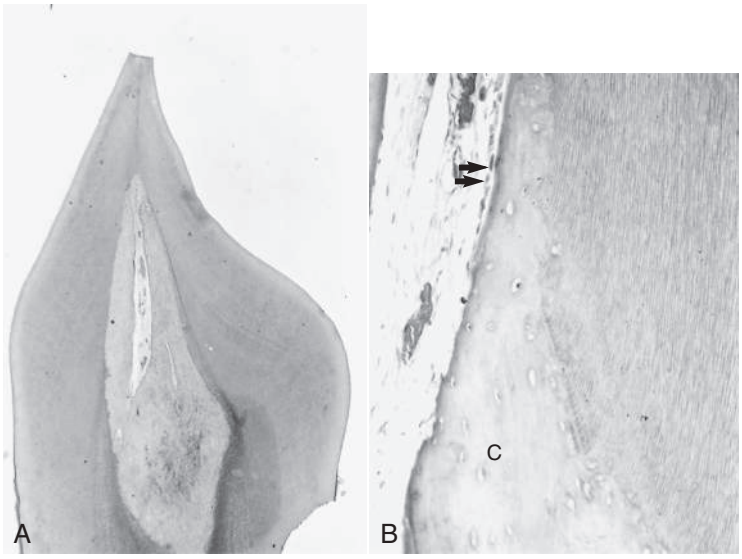


Fig. 13.55 **A,** Calcific metamorphosis of pulp tissue after luxation of tooth as a result of trauma. Note presence of soft-tissue inclusion. **B,** High-power view showing cementoblasts *(arrows)* lining cementum *(C)*, which has been deposited on the dentin walls.

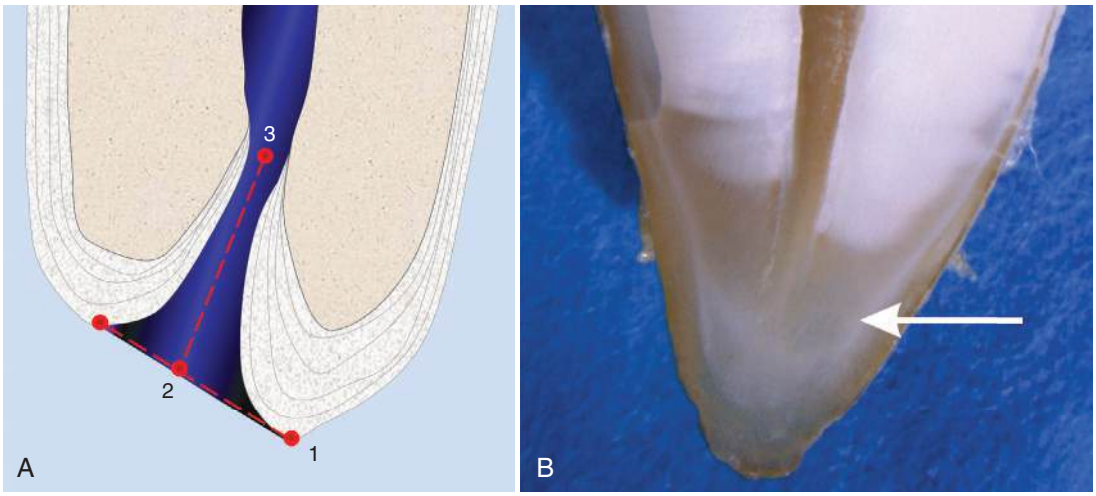


Fig. 13.56 Anatomical structures in the apical area. **A**, Cementum apposition during life will change the apical anatomical landmarks. Number *1* denotes the radiographic apex, *2* the apical foramen (major) and *3* the apical constriction (the minor apical foramen) representing the transition between the pulpal and periodontal tissues. The line that goes through *1* and *2* is often denoted the anatomical apex. **B**, Age-related sclerosis, making the dentinal tubules impermeable to bacteria, is a progressive event starting in the apical area *(white arrow)*.

pulpal infarction. The wide periapical foramen allows connective tissue from the periodontal ligament to proliferate and replace the infarcted tissue, bringing with it cementoprogenitor and osteoprogenitor cells capable of differentiating into either cementoblasts or osteoblasts or both.

When calcific metamorphosis is noted on a patient's radiograph, it is sometimes suggested that the tooth be treated endodontically because the pulp is expected to be secondarily infected, and endodontic therapy should be performed while the pulp canal is still large enough to instrument. In a classic study of luxated teeth, Andreasen[7] found that only 7% of the pulps that underwent calcific metamorphosis exhibited secondary infection. Because the success rate for nonsurgical endodontic therapy, not only in general[406] but also for obliterated teeth,[68] is considered high, prophylactic intervention does not seem to be warranted.

Age Changes

Continued formation of secondary dentin throughout life gradually reduces the size of the pulp chamber and root canals, although the width of the cementodentinal junction appears to stay relatively the same.[111,349] However, due to continuous cement application, the distance from the apical constriction to the radiographic apex may increase (Fig. 13.56).[388] The progressive sclerosis of dentin starting apically also has endodontic implications, as it makes dentine almost impermeable to bacterial invasion (see Fig. 13.56).[347] In addition, certain regressive changes in the pulp appear to be related to the aging process. There is a gradual decrease in the cellularity and a concomitant increase in the number and thickness of collagen fibers, particularly in the radicular pulp. The thick collagen fibers may serve as foci for pulpal calcification (see Fig. 13.53). The odontoblasts decrease in size and number, and they may disappear altogether in certain areas of the pulp, particularly on the pulpal floor over the bifurcation or trifurcation areas of multirooted teeth.

With age there is a progressive reduction in the number of nerves[104] and blood vessels.[23,25] Evidence also suggests that aging results in an increase in the resistance of pulp tissue to the action of proteolytic enzymes,[427] hyaluronidase, and sialidase,[25] suggesting an alteration of both collagen and proteoglycans in the pulps of older teeth. The main changes in dentin associated with aging are an increase in peritubular dentin, dentinal sclerosis, and the number of dead tracts.[¶,349] Dentinal sclerosis produces a gradual decrease in dentinal permeability as the dentinal tubules become progressively reduced in diameter.[357]

[¶]The term *dead tract* refers to a group of dentinal tubules in which odontoblast processes are absent. Dead tracts are easily recognized in ground sections because the empty tubules refract transmitted light, and the tract appears black in contrast to the light color of normal dentin.

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